When Background Noise Has Meaning

Environmental Sound Classification under Semantic Interference

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Abstract

Environmental recordings often contain more than one recognisable event. This study compares a handcrafted 92-feature SVM, a compact log-Mel CNN and frozen zero-shot LAION-CLAP on all five official ESC-50 folds. Identical white-noise, DEMAND-background and labelled event mixtures are tested at +10, +5 and 0 dB, with gain-matched controls and two activity-based power definitions. The SVM is especially vulnerable to white noise; CLAP retains stronger generic-background performance while frequently predicting a competing event. Temporal overlap changes the CNN comparison, and its negative semantic-minus-DEMAND contrast replicates directionally across three training seeds. In the expanded experiment, WGN augmentation raises three-seed CNN macro-F1 under +5-dB white noise from 7.48% to 43.95%, but changes same-category semantic performance only from 32.99% to 33.46%. Semantic augmentation reaches 36.05% in that condition, while clean macro-F1 falls from 65.78% to 61.70%; its modest direct CNN advantages over WGN training do not pass the declared multiplicity correction. A fixed prompt ensemble raises CLAP clean accuracy from 81.75% to 92.90% without changing audio embeddings, yet a substantial semantic-interference gap remains. Capture destinations, full-class rankings and within-model feature shifts expose failure structure that aggregate accuracy misses. At +5 dB, one-third of single-template CLAP captures retain the target at rank two. On small subsets matched closely on measured spectral and temporal proxies, CLAP retains a semantic-minus-DEMAND accuracy disadvantage while CNN intervals include zero; selection and unmeasured acoustics limit this interpretation. The contribution is a reproducible DSP experiment with measured robustness trade-offs, rather than an equal-data model contest or causal isolation of meaning from acoustics.

1 Research question and context

A single-label environmental-sound benchmark usually identifies a dominant event. Deployment audio can contain a dog bark and an alarm simultaneously. Equal interference power does not imply equal spectral structure, temporal overlap, recognisability or perceptual prominence. The main question is therefore: *How do different audio representations fail under ordinary acoustic noise and a competing meaningful event at controlled relative levels?* Two further questions ask whether an error identifies the distractor and whether ordinary noise augmentation transfers to competing meaningful events.

ESC-50 provides a small balanced benchmark with official folds [1, 2]. Spectrogram CNNs established a learned alternative to handcrafted descriptors [3]; augmentation has also been studied for environmental-sound classification [4]. A third paradigm, contrastive audio-language pretraining, permits classification through similarity to textual class descriptions [6]. These systems have very different external training histories. Their comparison here concerns deployment behaviour, not equal-data training efficiency.

Recent work motivates a closer examination of structured interference. PAT uses training-free cross-modal alignment and evaluates noisy inputs [7]. Mix-CLAP studies noise-aware audio-language models and efficiency [8]. RSA-Bench constructs complex acoustic backgrounds for speech perception and reasoning tasks [9]. That benchmark is related motivation, not a direct environmental-event classification baseline. This project does not reproduce those model-training programmes.

2 Dataset, models and validation

2.1 Official folds and data handling

ESC-50 contains 2,000 five-second recordings, 50 balanced classes and five predefined folds. All 50 classes are retained. For outer fold k, fold $(k \bmod 5) + 1$ supplies validation data and the remaining three folds supply selection training data. After selecting hyperparameters or training duration, a fresh model is fitted on the four non-test folds. Test labels are used only for evaluation. Feature standardisation and Mel normalisation are estimated from the applicable training partition.

The data audit identifies four source IDs spanning official folds, affecting nine target clips. Official assignments are preserved and documented; those sources are excluded from semantic-distractor eligibility. Consequently, the experiment should not be described as perfectly source-independent. The dataset authors also document partly class-dependent preprocessing [2]; clean benchmark performance is not evidence of unrestricted real-world generalisation.

2.2 DSP and classical baseline

The custom STFT uses a 1,024-sample periodic Hann window, a 512-sample hop and a 1,024-point FFT at 44.1 kHz. Frames start at sample zero; only the final incomplete frame is zero-padded. The output contains 513 complex one-sided bins. No per-clip amplitude normalisation or pre-emphasis is applied. Agreement is checked against SciPy and against an actual MATLAB execution of the unmodified course `MyStft.m`. The latter comparison explicitly reconciles the supplied window, leading padding, frame offsets and full/one-sided spectra. On a real ESC-50 recording the relative complex-valued error is approximately 2.6×10^{-16}. This verifies corresponding frames, not identical boundary policies or default window helpers.

The handcrafted representation aggregates means and standard deviations of RMS, zero-crossing rate, spectral centroid, spread, roll-off, flux, 20 MFCCs and their deltas, giving 92 dimensions. An RBF SVM uses training-only standardisation. The original nine-candidate grid combines $C \in \{1, 10, 100\}$ with $\gamma \in \{\texttt{scale}, 0.001, 0.1\}$, selected by clean validation macro-F1. The implementation retains one-vs-rest decision scores and explicit tie handling.

2.3 Compact CNN and frozen CLAP

The CNN uses 64 Mel bands, log compression, and three convolution/normalisation/ ReLU/pooling blocks with 16, 32 and 64 channels. Temporal pooling and four frequency bins yield a 256-dimensional representation before the classifier. It has 36,258 trainable parameters. Adam uses learning rate 10^{-3}, weight decay 10^{-4}, batch size 32 and dropout 0.3. Training is capped at 40 epochs with patience 8; clean validation selects duration before a fresh four-fold refit. The core experiment uses seed 5305. Additional seeds are analysed separately.

CLAP uses the fixed `630k-audioset-best.pt` checkpoint, HTSAT-tiny audio encoder, RoBERTa text encoder and no fusion. Audio is resampled to 48 kHz and passes through the official int16 truncation-and-return convention exactly once. That conversion clips to $[-1, 1]$ after resampling. The mixer's no-hard-clipping guarantee concerns the canonical input waveform before model-specific preprocessing. The official non-fusion frontend repeat-pads five-second inputs to ten seconds; all systems receive the same canonical five-second waveform before their own frontends. The prompt is `the sound of {label}`, replacing underscores with spaces. Class scores are cosine similarities between unit audio embeddings and the 50 unit prompt embeddings. Parameters are frozen; there is no ESC-50 classifier training, prompt selection or fine-tuning. Checkpoint SHA-256 starts `8053c977`; the complete hash is stored with the results. External pretraining overlap cannot be ruled out merely because inference is zero-shot.

3 Controlled interference and the power-definition change

Each target receives white Gaussian noise, a five-second segment from one of three DEMAND recordings (traffic, station and metro) [5], one same-coarse-category ESC-50 distractor and one different-coarse-category distractor. Semantic distractors have a different fine class and source recording and come from the target's held-out fold. Pairing is deterministic. Within each target/condition, the exact noise waveform, DEMAND segment or semantic pair is fixed across the three levels. Semantic events use zero relative offset. DEMAND is a real-background control, not a guarantee of semantically empty audio: traffic or station recordings can contain recognisable events. The semantic-distractor condition adds a known competing ESC-50 class, enabling label-specific capture analysis; it does not isolate meaning from every acoustic difference between conditions.

Let $x[n]$ be the target and $d[n]$ the interferer. Activity masks contain samples covered by unwindowed frames whose mean-square energy is within 40 dB of the corresponding peak frame. Define $P(z; M) = |M|^{-1} \sum_{n \in M} z[n]^2$. For requested level L,

$$P_x = P(x; M_x), \tag{1}$$

$$P_d^{\text{own}} = P(d; M_d), \qquad P_d^{\text{target}} = P(d; M_x), \tag{2}$$

$$\alpha = \sqrt{P_x / (10^{L/10} P_d)}, \qquad y[n] = g\big(x[n] + \alpha d[n]\big), \tag{3}$$

$$g = \min\left(1, \frac{0.99}{\max_n |x[n] + \alpha d[n]|}\right). \tag{4}$$

The matching clean control is $gx[n]$. Common attenuation preserves the component-power ratio and avoids hard clipping. Achieved ratios are recomputed numerically rather than assumed from the gain formula.

Deviation from the proposal. The proposal specified the target-mask definition. In preliminary implementation, sparse distractors with little energy inside M_x required very large gains, even when their event occurred outside the target-active interval. The default was therefore changed to own-activity power. The originally proposed target-mask

protocol was also run in full as a sensitivity analysis. The two ratios answer different measurement questions; neither is an automatic correction for temporal overlap, and the choice is not attributed to an unrecorded instructor instruction.

There are 24,000 own-protocol mixtures. Three target/distractor pairs have zero interferer energy on the target mask; their three levels are undefined under the target protocol and are excluded with recorded reasons, leaving 23,991 mixtures. Cross-protocol comparisons use common retained targets. The manifest records target/distractor identity, labels, categories, folds, hashes, seeds, offsets, gains, both measured power ratios, peak control and activity overlap $|M_x \cap M_d|/|M_x|$. The 0-dB condition is a deliberately ambiguous single-label stress test, not an ordinary equal-difficulty accuracy benchmark.

4 Outcomes and statistical treatment

Accuracy and macro-F1 are computed within each official test fold, with means and sample standard deviations across five folds. Macro-F1 retains all 50 class labels. For a semantic condition, distractor capture rate is the number of wrong predictions equal to the distractor divided by the number of errors. The unconditional capture fraction uses all trials instead. A reference of 1/49 assumes uniform choice among wrong labels; it is not an empirical null for class-dependent confusion.

The target–distractor margin subtracts the distractor score from the target score. Scores are SVM decision values, CNN logits or CLAP cosines and are interpreted only within a system. A negative margin does not guarantee that the distractor beats every other class. The CLAP diagnostic therefore also computes ranks and competing scores across all 50 prompts.

Relative macro-F1 loss is $1 - F_{\text{mixture}}/F_{\text{clean}}$, where each F is the five-fold mean on identical retained targets. It is a proportional loss, not a percentage-point difference. Replacing the reference with $F_{\text{gain control}}$ gives the additional control-referenced loss. Capture rates pool counts across folds; they are not unweighted means of fold capture percentages. Fold SD describes between-fold variation, not a confidence interval for a population mean.

Paired bootstrap intervals resample target clips within official folds, retaining matched condition predictions, using 2,000 replicates and seed 530509. For these matched comparisons, accuracy differences pool the retained target trials; macro-F1 differences average the fold-specific values. They condition on the fitted models and do not incorporate refitting or all source-recording dependence. Exact McNemar comparisons use paired correctness. Holm correction covers the 12 +5-dB high-overlap comparisons across models, power definitions and semantic pairing types. These are post-baseline extensions, not retrospectively preregistered tests. Three CNN seeds separately assess training variability; five folds times three seeds are not 15 independent datasets.

Three comparison families are kept distinct: 12 core high-overlap contrasts (three systems, two power protocols and two semantic pairing types); 12 clean-CNN replication contrasts (three seeds, two protocols and two pairing types); and 156 intervention contrasts, defined in the extension protocol. A core seed-5305 comparison appears in both of the first two families, so its adjusted p value can differ without its effect or raw p value changing. Holm correction applies within each family; it is not a study-wide guarantee across every exploratory analysis. Bootstrap intervals are pointwise and unadjusted. Exact McNemar tests assume independent target pairs under the null; shared source recordings, distractors, backgrounds and overlapping training sets limit that assumption. The corrected values therefore remain conditional, nominal tests, rather than proof of generalisation to independently sampled acoustic scenes.

An exploratory resampling-unit sensitivity analysis draws 10,000 bootstrap samples of original target source IDs globally. All retained clips from a sampled source move together, including the rare source spanning official folds. It estimates the pooled matched-target accuracy difference. These pointwise source- cluster intervals condition on fitted models, do not replace the declared Holm families, and do not capture every dependency from shared distractors or backgrounds.

5 Core results

Table 1: Clean five-fold results (percent; mean ± sample SD). Core CNN seed: 5305.

System	Accuracy	Macro-F1
SVM	48.90 ± 4.21	47.96 ± 4.07
CNN	67.90 ± 3.65	66.61 ± 3.91
CLAP	81.75 ± 2.58	79.49 ± 3.28

Table 1 establishes the three clean reference points. Table 2 shows +5-dB macro-F1 for each power definition; the complete level curves are in Figure 1. The SVM is severely affected by white noise, whereas meaningful distractors are

Table 2: Macro-F1 at +5 dB (percent; five-fold mean ± sample SD). Each cell uses its retained targets.

Power	Condition	SVM	CNN	CLAP
Own	White noise	2.03 ± 0.86	7.76 ± 1.54	62.91 ± 3.50
Own	DEMAND	16.13 ± 1.95	29.46 ± 4.19	71.49 ± 1.00
Own	Same category	22.47 ± 2.68	32.60 ± 2.26	48.86 ± 1.53
Own	Different category	21.34 ± 2.58	32.00 ± 3.38	45.16 ± 1.11
Target	White noise	2.03 ± 0.86	7.76 ± 1.54	62.92 ± 3.50
Target	DEMAND	16.15 ± 2.29	29.33 ± 4.14	71.64 ± 0.91
Target	Same category	21.14 ± 2.63	31.70 ± 1.88	47.76 ± 1.07
Target	Different category	20.87 ± 2.37	30.93 ± 4.19	42.81 ± 1.27

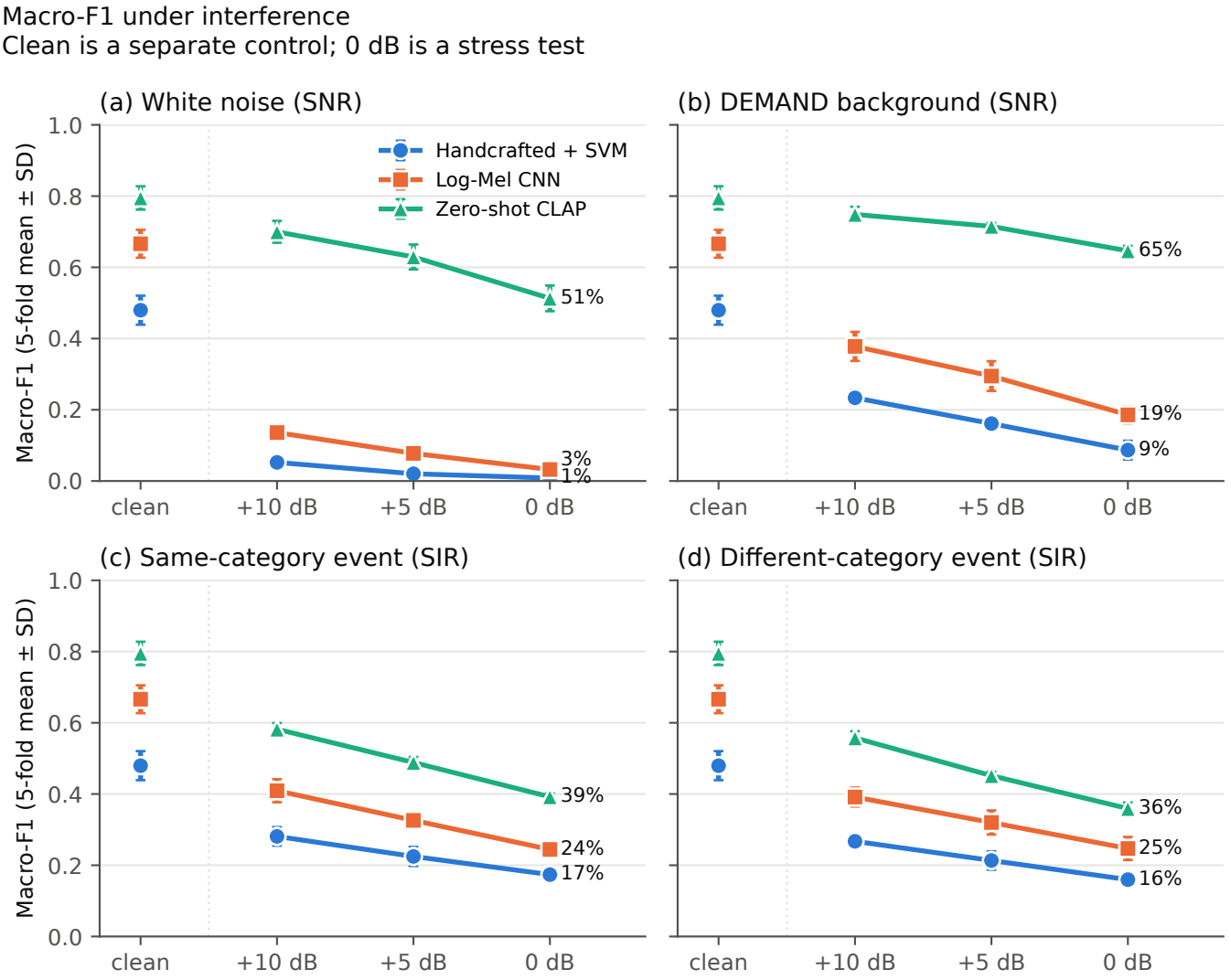


Figure 1: Core robustness curves under own-activity power. Values are five-fold macro-F1 means; error bars are sample SD, not confidence intervals. The same input mixtures are used by all systems. Zero dB is the equal-power stress test.

less damaging in its aggregate results. This contradicts a universal assertion that semantic interference must always be harder than generic noise. CLAP is comparatively robust to ordinary backgrounds but experiences a much larger semantic-interference penalty.

Under the own protocol at +5 dB, CLAP's capture rates are approximately 67% for same-category and 60% for different-category distractors. The corresponding CNN values are about 31% and 28%, and SVM values about 19% and 16%. These are fractions of errors, not percentages of all targets. Capture and margin curves describe increasingly strong distractor competition as SIR falls (Figures 2 and 14). Both absolute accuracy and capture denominators must accompany this interpretation.

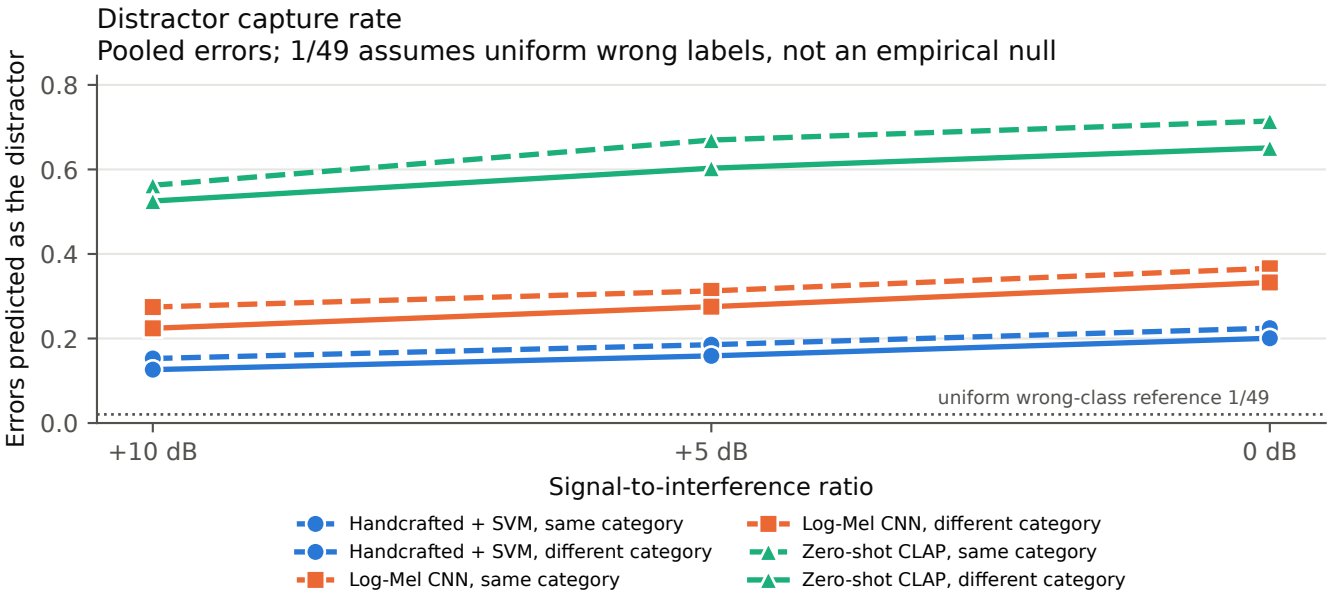


Figure 2: Distractor capture among incorrect trials. The horizontal reference is uniform wrong-label choice, not a class-conditioned null distribution. Same-category and different-category pairings are distinguished by line style.

5.1 Overlap and matched comparisons

The whole-dataset CNN ranking between semantic interference and DEMAND is affected by the composition of low-overlap mixtures. Compare each semantic condition against DEMAND on precisely the same targets, stratified by overlap. At overlap at least 0.8 and +5 dB, CNN semantic accuracy is about three percentage points lower. Pointwise paired bootstrap intervals exclude zero, but the Holm-adjusted exact McNemar p-values are approximately 0.12–0.15. The appropriate claim is a consistent observed direction requiring replication, rather than an established family-wise significant CNN effect.

CLAP's corresponding losses are much larger: own-protocol differences are -25.55 points (same category) and -28.53 points (different category), with paired 95% intervals [-28.28,-22.95] and [-31.20,-25.73]. The target protocol gives similarly large differences. Figure 3 separates these effects from the small, uncertain SVM differences. Stratified macro-F1 is also reported, but it can be depressed by missing classes; matched accuracy and class coverage are used to interpret stratum difficulty.

Changing power convention moderately lowers semantic performance but does not remove the main model-specific pattern. It should not be claimed that every fine ranking is invariant: small same-/different-category order changes can occur, including for the CNN at 0 dB. Both protocols and the retained-target intersection remain part of the result rather than selecting a preferred table.

5.2 Gain-matched control check

Table 3 compares attenuated and original clean audio on matched targets. The shifts are small relative to mixture losses, but are not zero. Across all 72 core cells, control-minus-clean F1 ranges from -1.74 to +0.49 percentage points.

Table 3: Gain-matched clean minus original clean macro-F1 at +5 dB, in percentage points. Each value is a difference of five-fold means on that condition's retained targets. Core CNN seed: 5305.

Power	Model	White	DEMAND	Same	Different
Own	SVM	-0.25	-0.19	-0.64	-0.76
Own	CNN	+0.28	+0.25	-0.10	+0.03
Own	CLAP	+0.25	+0.15	+0.11	+0.45
Target	SVM	-0.25	-0.22	-0.47	-0.48
Target	CNN	+0.28	+0.25	-0.35	-0.32
Target	CLAP	+0.25	+0.15	+0.40	+0.13

Both reference losses are exported; these results show limited attenuation effects, not exact amplitude invariance.

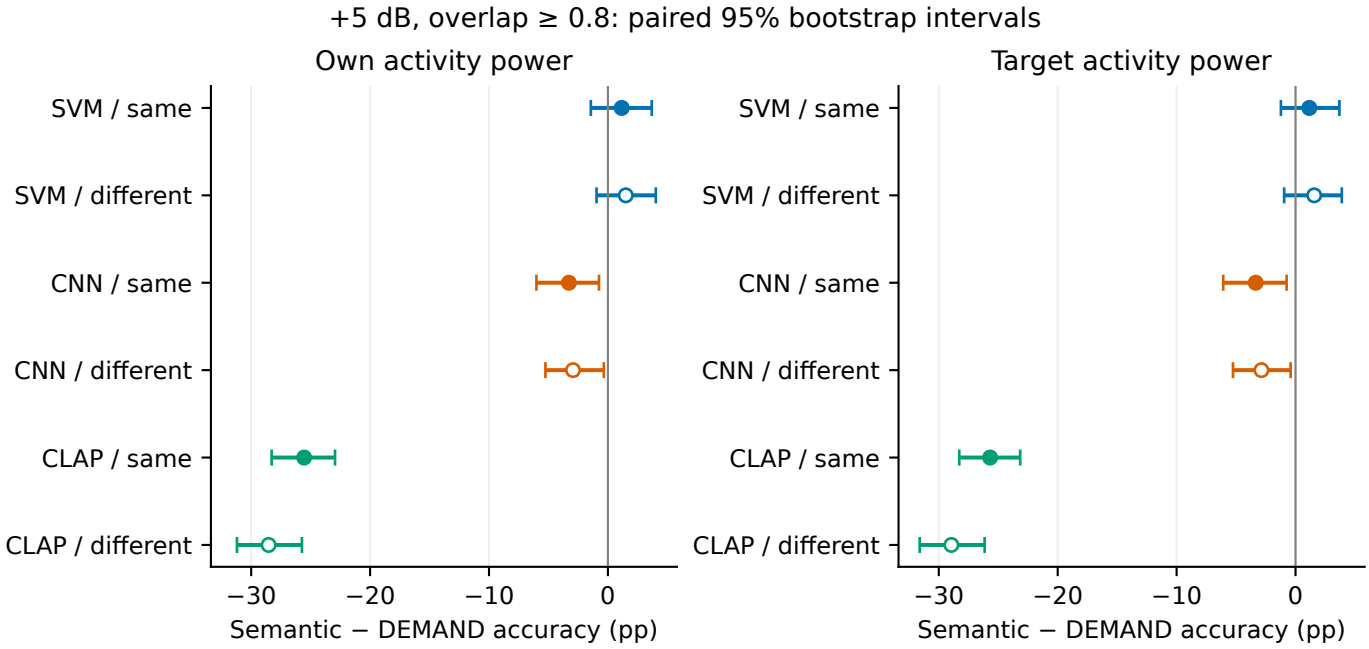


Figure 3: Semantic-minus-DEMAND accuracy on matched high-overlap targets at +5 dB. Intervals are 2,000-replicate paired target-level bootstrap intervals within folds, conditional on the trained models. They are pointwise intervals; the text separately reports multiplicity-adjusted tests.

6 Representation and competition diagnostics

Handcrafted displacement uses the relevant SVM training-fold scale and is normalised by the square root of group dimensionality. The eight groups are RMS, ZCR, centroid, spread, roll-off, flux, MFCCs and delta-MFCCs. At +5 dB, white noise shifts ZCR and spectral shape descriptors far more than the other conditions (Figure 4). This offers a concrete DSP explanation for the SVM's extreme white-noise sensitivity without treating feature movement as sufficient to explain every wrong label.

CNN and CLAP movement uses cosine distance between mixture and gain-matched clean embeddings. Distances are

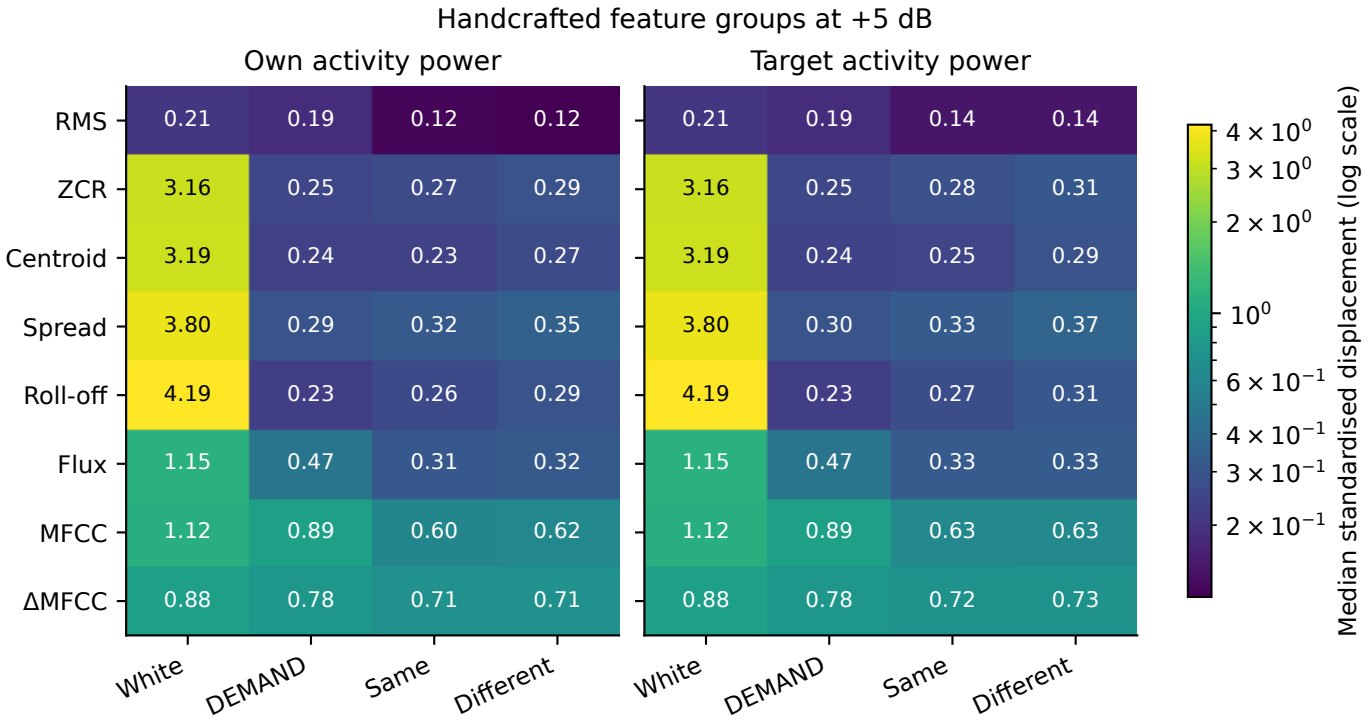


Figure 4: Median clean-control-to-mixture displacement for eight handcrafted feature groups at +5 dB. Each group uses training-fold standardisation and division by the square root of its dimension. The shared logarithmic colour scale is a display scale; annotated values are untransformed distances.

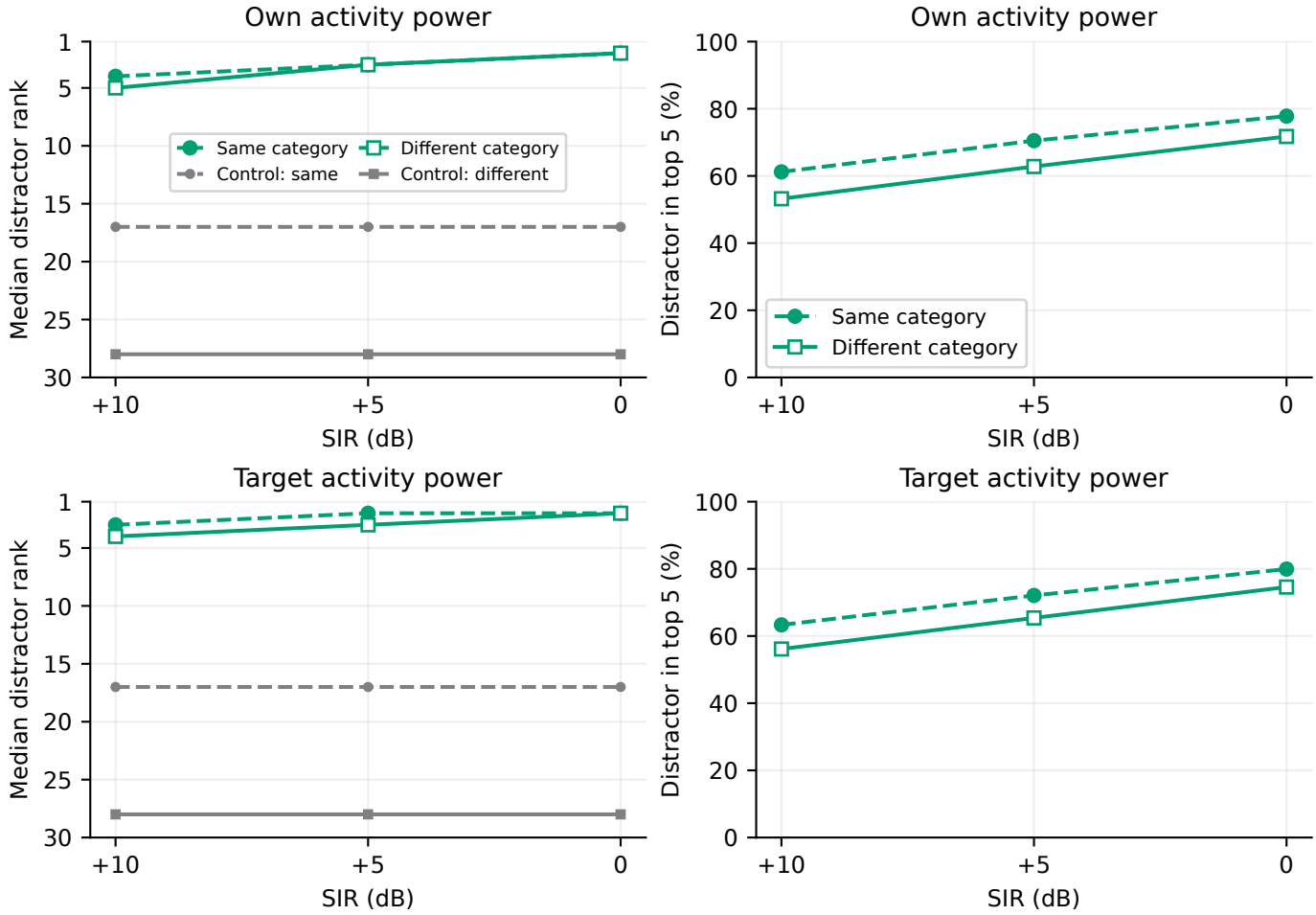


Figure 5: CLAP distractor ranking among all 50 classes. Lower rank is stronger; rank 1 is highest. Top-five fractions use all semantic trials, unlike error-conditioned capture rate. Grey curves are gain-matched clean controls.

not compared absolutely across architectures. For high-overlap CNN pairs, semantic interference can produce worse accuracy despite smaller movement than matched DEMAND backgrounds. This is compatible with direction-sensitive decision boundaries, but does not prove that semantic meaning itself caused the effect. CLAP behaves differently: its high-overlap semantic mixtures generally move farther than the matched DEMAND embeddings.

CLAP also exposes the destination of movement. Under own-activity power, the same-category distractor's median rank improves from 17 in the clean control to 4, 3 and 2 at +10, +5 and 0 dB. Figure 5 shows the complete 50-class ranking diagnostic. Scores for the three highest other competing classes are retained per trial. Displacement/error AUC (Appendix) summarises association within each condition and fold, not causal explanation.

6.1 The remaining competing classes

An error need not name the designated distractor. For each CLAP mixture, define $o^* = \arg\max_{c \notin \{x,d\}} s_c$ over the other 48 labels. Table 4 retains all three possible prediction destinations and shows the target's margin over this strongest other class. The last column subtracts that same class's gain-matched control score from its mixture score. Because o^* is selected on the mixture, this is a descriptive selected-competitor diagnostic; it is not evidence for a prespecified class effect.

Table 4: Single-prompt CLAP at +5 dB. Target, distractor and other columns are disjoint prediction counts summing to n. The final two columns are median cosine-score differences, not probabilities.

Power	Pair	n	Target	Distractor	Other	$s_x - s_{o^*}$	Δs_{o^*}
Own	Same	2000	1056	632	312	+0.062	+0.047
Own	Different	2000	985	612	403	+0.044	+0.055
Target	Same	1998	1032	664	302	+0.057	+0.053
Target	Different	1999	943	647	409	+0.036	+0.062

6.2 Closing analysis: a ranked two-label readout

These final analyses use existing predictions and audio only. They are exploratory: the protocol was fixed after inspecting the cloud top-two and partial acoustic results, before the local DEMAND completion. No model, prompt or dataset was added. The original single-label results remain unchanged.

An argmax capture does not establish that the target has disappeared from the ranking. We therefore select exactly the top k labels, with $k = 2$ primary and $k \in \{1, 3, 5\}$ as sensitivity checks. Equal scores use ascending class ID; no ties occurred at the inspected boundaries. Both the target and distractor ranks are recovered from all 50 saved scores, including gain-matched clean controls. For each designated pair the four states are target only, both, distractor only and neither. Fractions use all semantic trials, whereas target retention among captures conditions on the original prediction equalling the distractor. There are 23,991 semantic trials per system or text-readout variant across both power definitions. This is ranked retrieval, not a calibrated multi-label detector: two labels are always returned, and a rank does not demonstrate subjective hearing or complete source recovery.

Table 5: CLAP top-two states at +5 dB (percent of all semantic trials). Last column: target in top two among original argmax captures, a different denominator. Own uses 2,000 targets per type; target uses 1,998 same-category and 1,999 different-category targets.

Power	Text readout	Type	Target only	Both	Distractor only	Neither	Retained
Own	Single	Same	43.95	26.05	23.35	6.65	37.34
Own	Single	Different	44.60	21.00	24.05	10.35	33.17
Own	Ensemble	Same	41.55	36.65	19.05	2.75	47.60
Own	Ensemble	Different	45.55	28.45	22.90	3.10	38.60
Target	Single	Same	42.09	26.73	24.72	6.46	36.45
Target	Single	Different	41.97	22.16	25.66	10.21	32.92
Target	Ensemble	Same	38.99	38.34	19.92	2.75	47.85
Target	Ensemble	Different	42.47	29.96	24.41	3.15	38.65

At +5 dB under own power, the single-template target remains second in 236/632 same-category captures (37.34%) and 203/612 different-category captures (33.17%). Joint top-two inclusion occurs in 26.05% and 21.00% of all mixtures, compared with 5.40% and 1.70% of their clean controls. Yet 23.35% and 24.05% contain only the distractor among the two designated labels. The existing prompt ensemble retains the target in 47.60% and 38.60% of captures. The target-power analysis gives a similar partial recovery (Table 5). Thus some single-label failures retain useful target ranking, while the top-two readout does not remove the interference problem. Increasing k mechanically makes inclusion easier; Figure 21 reports that trade-off without reclassifying the original errors as correct. Even the two designated source labels need not exhaust every audible event in a recording.

6.3 Closing analysis: spectral and temporal overlap

All 24,000 own-power and 23,991 target-power rows now have acoustic descriptors, including 6,000 DEMAND rows in each protocol. The legacy long-term dB Mel correlation, linear spectral cosine, target time–frequency masking fraction, energy-weighted masking fraction and temporal activity overlap are retained. We add a cosine between spectra averaged only over target-active frames and a dB correlation after dividing each Mel band by its summed filter weights. The latter is a bandwidth-normalised density diagnostic. All use the project's 64-band DSP/CNN frontend, not CLAP's internal frontend. A target-active frame has unwindowed energy within 40 dB of its peak; a target-active time–frequency cell is within 40 dB of peak Mel-cell power. The masked-cell fraction counts those cells where scaled interference power exceeds target power. It is a computational proxy, not a validated measure of perceptual masking. Undefined target-active cosines are left missing (nine own-power rows), not replaced by zero.

This frontend detail changes the interpretation of the preliminary analysis. At +5 dB the white-noise median raw dB Mel correlation is −0.728; after bandwidth normalisation it is −0.008 (Figure 7). White-noise power per unequal-width Mel band is not flat, even when its power density is flat. The apparent anticorrelation is therefore not evidence that white noise physically avoids the target spectrum. Median masked-cell fractions are 0.319 for white noise, 0.146 for DEMAND, 0.121 for same-category events and 0.120 for different-category events under own power. These marginal summaries describe different spectral and temporal structures, not equivalent acoustic difficulty or semantic causality.

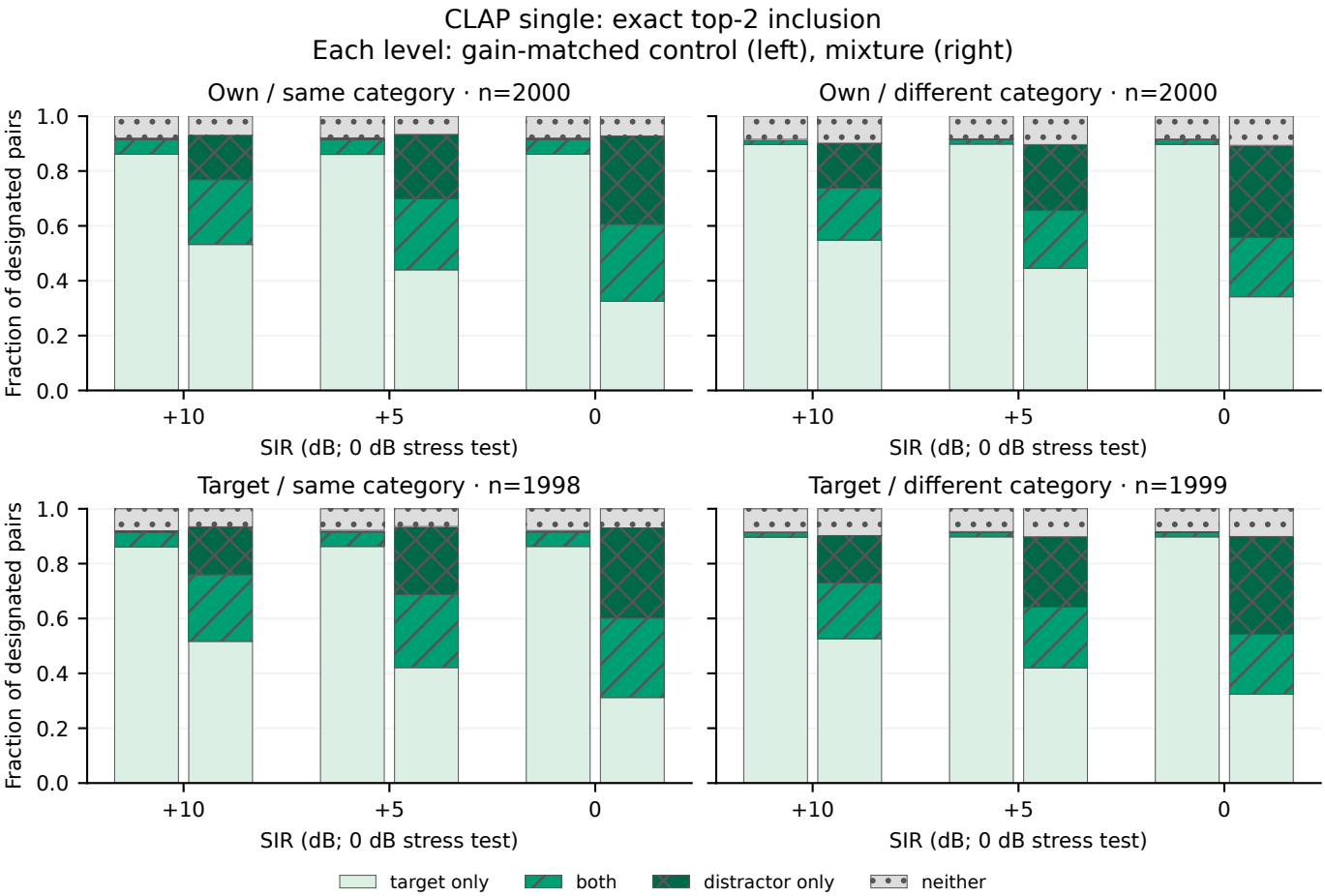


Figure 6: Exact top-two states for frozen single-template CLAP. Each SIR has gain-matched control (left) and mixture (right). Four states partition every designated target–distractor pair; the states are not perceptual detection claims. Counts and all-system sensitivities accompany the figure.

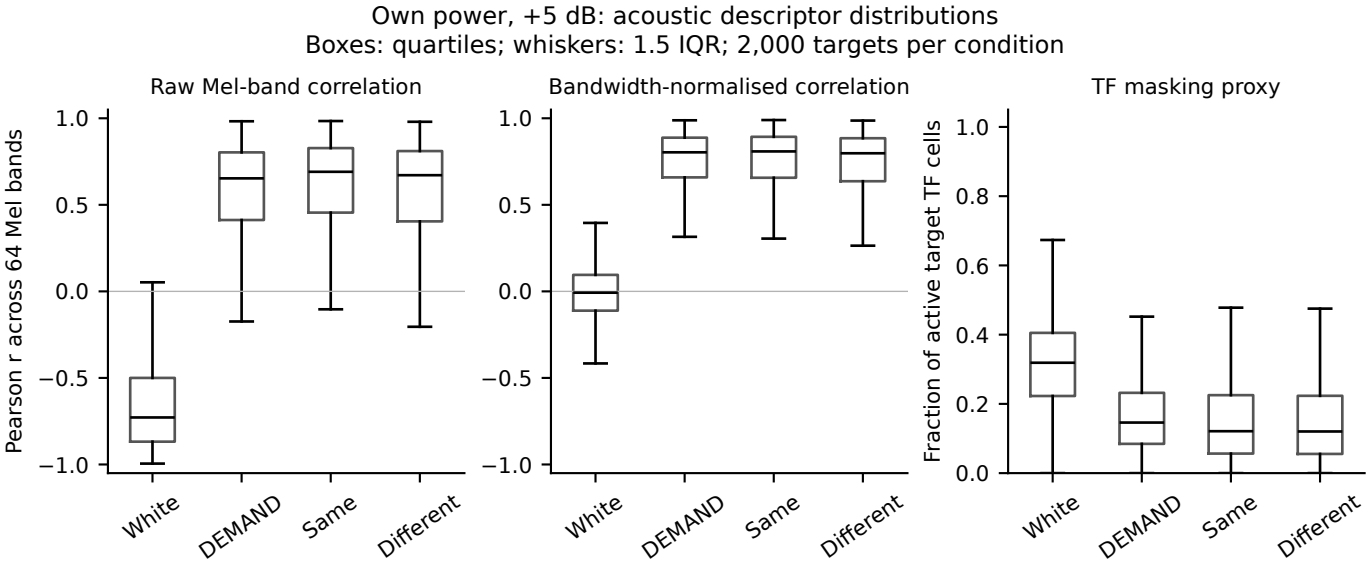


Figure 7: Acoustic descriptor distributions at +5 dB, own-power protocol. Each box represents 2,000 target mixtures; whiskers extend to 1.5 interquartile ranges. Bandwidth normalisation changes the white-noise correlation interpretation. Masking is a time–frequency proxy; each descriptor has its own scale. Target-power distributions are exported separately.

6.4 Same-target comparisons with close acoustic descriptors

The primary +5-dB comparison retains a target only when its two mixtures differ by at most 0.10 in target-active spectral cosine, 0.05 in masked-cell fraction, and 0.10 in temporal overlap. Selection uses audio descriptors alone and is identical for SVM, core CNN seed 5305 and single-template CLAP. Strict $(0.05, 0.025, 0.05)$ and loose $(0.20, 0.10, 0.20)$ versions accompany it, as do all-target and high-temporal-overlap comparisons. These calipers improve agreement on measured descriptors; they do not make the waveforms identical.

The primary same-/different-category versus DEMAND subsets retain 210/178 targets under own power (199/168 source clusters; 49/48 classes) and 205/187 under target power (195/178 clusters; 49/47 classes). Only about 9–10.5% of the benchmark is retained. Accuracy differences compare the same targets within each selected subset. Pointwise 95% percentile intervals use 2,000 bootstrap resamples of original target-recording IDs, keeping clips from one source together (seed 530510). They are conditional on the fitted models and observed pairings; shared interferers, backgrounds and training-set dependence are not fully captured. No new family of significance tests is introduced. Intervals are omitted when either target or source-cluster count is below 30.

Table 6: Accuracy difference in the primary acoustic-closeness subsets at +5 dB: semantic minus DEMAND, percentage points [pointwise 95% source-bootstrap interval]. Positive favours the semantic mixture. Each column compares identical targets within its own subset; subset sizes are given in the text.

Power	Model	Same category minus DEMAND	Different category minus DEMAND
Own	SVM	+1.90 [-4.42, +8.13]	+8.43 [+2.78, +14.61]
Own	CNN	+3.81 [-3.33, +11.79]	+2.25 [-3.37, +7.96]
Own	CLAP	-25.71 [-32.71, -19.05]	-19.10 [-26.74, -11.80]
Target	SVM	+2.44 [-3.81, +8.96]	+8.56 [+3.21, +14.05]
Target	CNN	+3.90 [-3.94, +11.77]	+3.74 [-2.12, +9.19]
Target	CLAP	-23.41 [-30.30, -16.19]	-16.04 [-23.28, -8.95]

CLAP's semantic disadvantage remains about 16–26 percentage points in these selected populations (Table 6); its strict and loose semantic-minus-DEMAND intervals also remain below zero. Measured spectral and temporal proxies therefore do not eliminate the observed CLAP gap on their available common support. This is residual association, not proof of an independent causal effect of meaning: onset detail, modulation, event recognisability and salience remain uncontrolled, and DEMAND itself can contain meaningful events.

The CNN point estimates in the close-descriptor subsets are instead positive, with all four intervals including zero. Its original high-overlap semantic disadvantage is still present in the earlier, larger population. Because target composition changes, the difference between those results cannot quantify how much acoustics explains. SVM same-category intervals also include zero; different-category contrasts remain positive. Direct same-minus-different contrasts on their own primary matched subsets have intervals spanning zero for all three models in both power protocols. They do not establish an independent coarse-category effect.

Descriptive bin summaries and logistic fits remain available as secondary diagnostics. In particular, a change in a logistic condition coefficient after adding covariates is not a percentage of the effect explained: odds ratios are non-collapsible, and support and model specification also matter [10]. In-sample pseudo-R^2 is not explained causal variance. The middle third of a masking-difference distribution is labelled as a third, not as an acoustically matched group. Accordingly, the regressions are descriptive sensitivity checks and do not separate semantic from acoustic causes.

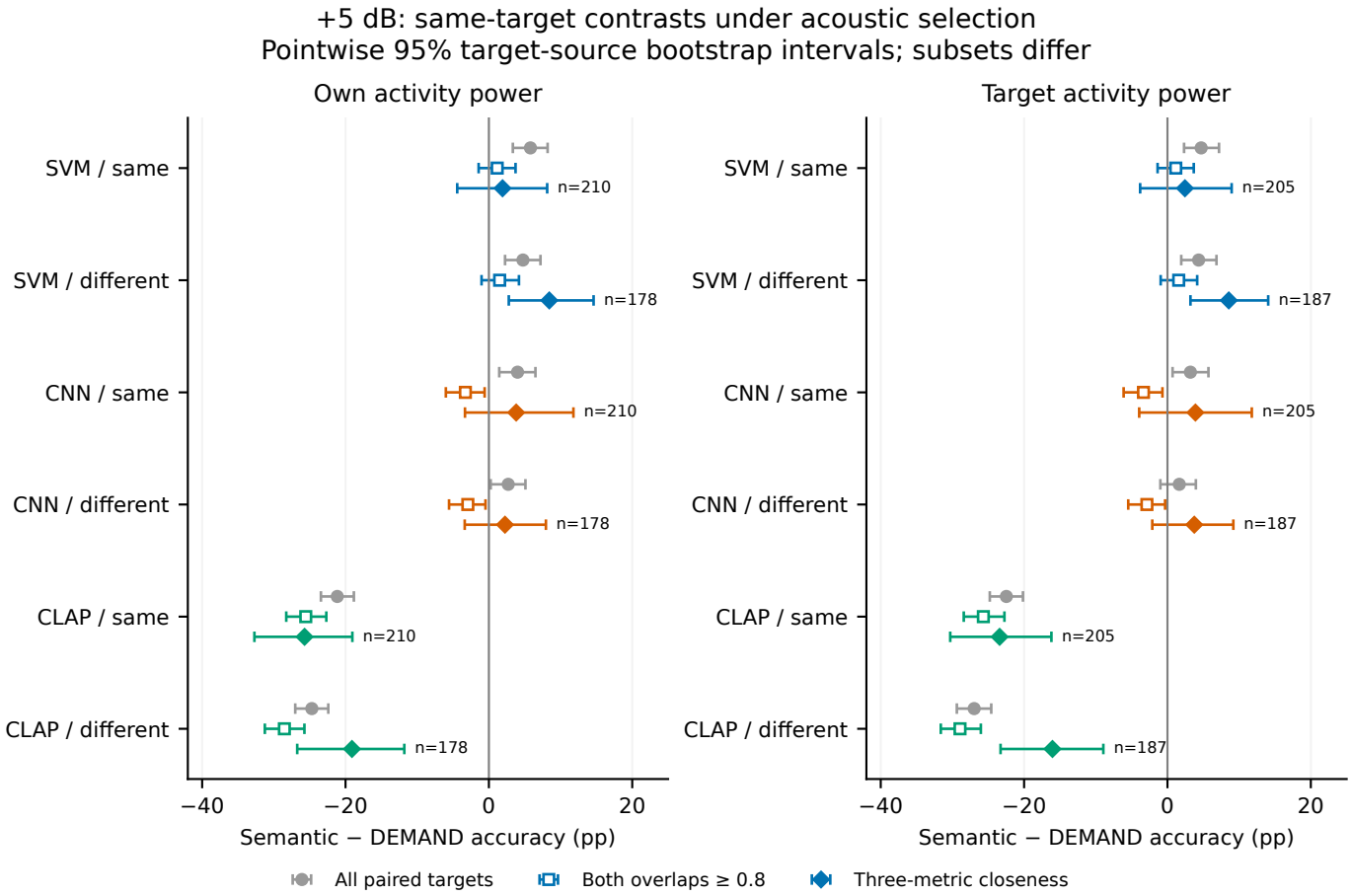


Figure 8: Same-target semantic-minus-DEMAND accuracy at +5 dB. Grey: all eligible targets; open markers: both mixtures have temporal overlap at least 0.8; filled markers: primary acoustic-closeness subset. Error bars are pointwise 95% target-source bootstrap intervals; displayed counts refer only to the close subset. The three selections are different populations, not sequential estimates of a causal mediation effect.

7 Deployment cost and limits

The expanded benchmark uses the same Apple M3 CPU, one numerical thread and batch size one for seven model variants, after project training and evaluation have stopped. One deterministic recording from each class and official fold gives 250 unique five-second clips; three repetitions give 750 timed queries per model in each pass. After observing different frontend times for CNNs with identical processing, a post-hoc second pass reverses the model order. Both passes are retained, giving 10,500 timed queries in total. Model, audio, software and benchmark-source hashes match across passes, and all timed predictions match the formal clean evaluations. Ten warm-ups precede each fold's panel. Total latency directly times preprocessing, representation extraction, all-class scoring and argmax. Disk I/O, model loading and one-off text preparation are excluded. Model-specific pinned software environments are recorded. CNN latency uses seed 5305; its three-seed accuracy analysis is separate. This is a two-order descriptive sensitivity on one machine, not a randomised hardware study. The timing shift cannot be assigned to a measured thermal or scheduling cause. In particular, differences among identically shaped CNNs are not evidence that augmentation changes operation count.

Table 7: Balanced CPU panel: both timing passes. Medians and p95 values are in milliseconds. Serialized size is one fold's model (median across folds), or the CLAP checkpoint; it is not resident memory.

Model / training	Median 1	Median 2	p95 1	p95 2	File (MB)
SVM / clean	15.00	11.29	22.01	11.79	2.41
SVM / WGN	14.95	11.47	26.39	12.11	4.83
SVM / semantic	13.92	11.42	15.69	12.19	4.68
CNN / clean	3.07	2.44	3.61	2.59	0.16
CNN / WGN	4.83	2.43	6.42	2.64	0.16
CNN / semantic	2.97	2.48	3.17	2.68	0.16
CLAP / single prompt	69.30	50.86	97.44	53.04	1863.59

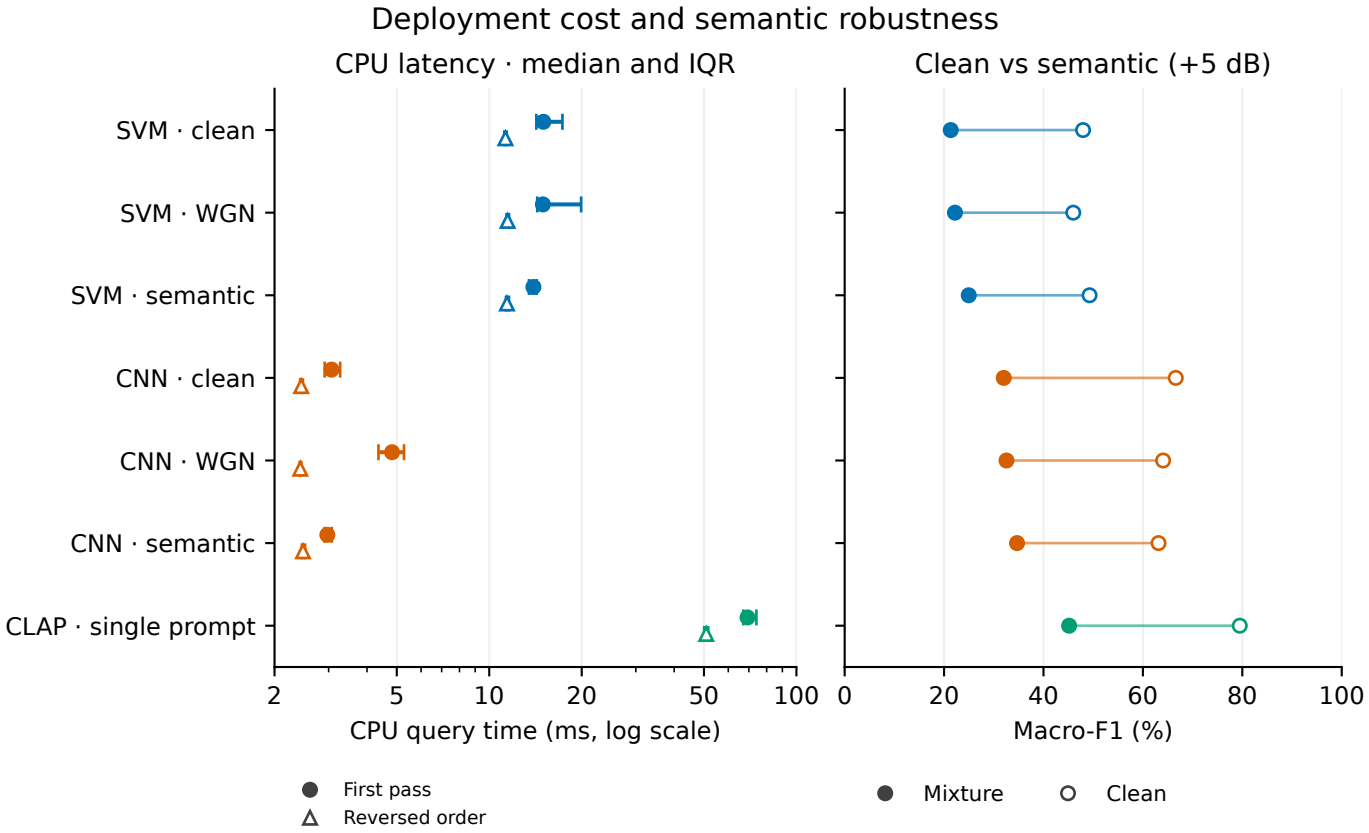


Figure 9: Balanced CPU latency alongside clean and +5-dB different-category macro-F1 under own-activity power. Both timing passes are retained; whiskers show each pass's interquartile range of 750 queries, not confidence intervals. CNN performance here uses seed 5305; the intervention analysis retains all three seeds.

External pretraining is excluded. CLAP prompt embeddings are cached, and its audio embedding plus class scoring is included. CPU performance is separate from MPS used for supervised training. CLAP's reported preprocessing stage covers resampling and quantisation; its internal Mel frontend is included in embedding time. Stage boundaries therefore differ across implementations, so total query cost is the comparable measure.

The compact CNN has 36,258 parameters; the loaded CLAP model has 158,334,169. Saved checkpoint size, parameter bytes and inference latency are different quantities. CLAP's 1.86-GB checkpoint must not be compared to a small supervised model file as though both contain identical kinds of state. The project measures local deployment cost, not total lifetime training energy.

Power matching alone leaves spectral, temporal and perceptual differences uncontrolled. The closing analysis measures several spectral and temporal proxies, but its restricted matching does not eliminate the remaining acoustic or perceptual differences. Only three DEMAND environments and one semantic pair of each category type per target are sampled. Coarse ESC categories are not a direct measure of perceptual similarity. At 0 dB either event may be a reasonable description. Frozen CLAP has external pretraining advantages and potential dataset overlap. Source reuse, post-baseline analysis choices and seed variability limit generalisation. These limitations are reasons to report the protocol and failure cases precisely, not reasons to reinterpret a smaller error bar as proof of causal semantics.

8 Extension experiments

8.1 Protocol

The extension compares clean, WGN-augmented and semantic-augmented supervised training. CNN seeds are 5305, 5306 and 5307. Each augmented CNN sees clean data with probability 0.5 and otherwise one of three fixed training-only views at +5, +10 or +15 dB under the own-activity power convention. Both power conventions are then evaluated at test time. Banks are shared across optimisation seeds. Clean validation, architecture, training cap and four-fold refit logic are unchanged. Actual refit epochs are selected separately on clean validation, so identical augmentation probabilities and maximum budgets do not imply identical numbers of training-example presentations. Semantic partners come only from the current training partition; test DEMAND recordings are never used for augmentation. The SVM uses equal counts of clean examples and one balanced fixed augmented view per training clip. These interventions test transfer

from ordinary acoustic corruption to competing events.

A separate fixed prompt-ensemble ablation averages text embeddings over the complete released PAT template list and reuses the exact cached CLAP audio embeddings. It does not select prompts using labels and is not called full PAT. The inspected reference implementation uses a different CLAP backend, local features and a label-based fusion-coefficient search; reproducing that search on outer test folds would invalidate this experiment's isolation. None of those steps is silently incorporated into the core CLAP reference.

8.2 Verified results

All declared intervention variants have completed evaluation. Additional clean runs are seed 5306: clean accuracy 66.40%, macro-F1 65.19%; seed 5307: clean accuracy 66.40%, macro-F1 65.53%.

CNN replication across optimisation seeds

All three clean-trained seeds show lower accuracy for semantic interference than matched DEMAND at +5 dB and overlap at least 0.8. The two new seeds have larger differences than the core seed. Their eight comparisons pass the declared 12-comparison Holm family, while seed 5305 remains above 0.05. This supports directional replication; it does not turn repeated predictions of the same clips into independent samples or isolate a causal semantic effect.

Table 8: Semantic minus matched DEMAND accuracy (percentage points), with paired 95% target-bootstrap intervals and Holm-adjusted exact McNemar p values. Each seed is analysed separately.

Power	Seed	Same category	p	Different category	p
Own	5305	-3.28 $[-5.94, -0.68]$	0.060	-2.94 $[-5.46, -0.48]$	0.060
Own	5306	-5.74 $[-8.40, -3.14]$	< 0.001	-6.83 $[-9.28, -4.23]$	< 0.001
Own	5307	-5.12 $[-7.79, -2.46]$	0.001	-5.80 $[-8.33, -3.34]$	< 0.001
Target	5305	-3.35 $[-6.01, -0.68]$	0.060	-2.87 $[-5.46, -0.41]$	0.060
Target	5306	-5.74 $[-8.33, -3.14]$	< 0.001	-6.62 $[-9.15, -3.89]$	< 0.001
Target	5307	-4.92 $[-7.58, -2.25]$	0.002	-5.67 $[-8.19, -3.21]$	< 0.001

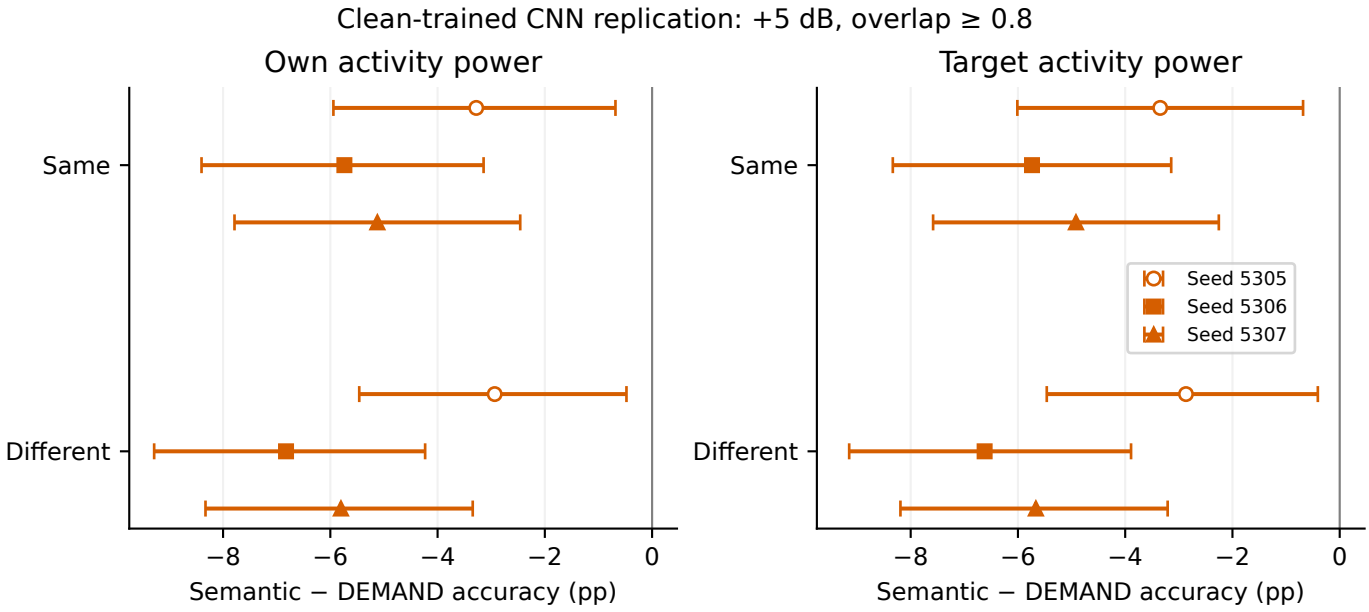


Figure 10: The high-overlap CNN contrast is negative for all three seeds in both power protocols. Pointwise 95% intervals use 2,000 within-fold target-bootstrap replicates and 1,464 same-category or 1,465 different-category paired targets. Intervals condition on each fitted model; corrected tests are reported separately in the table.

The exploratory target-source bootstrap preserves this direction: all 12 clean-CNN pointwise source-cluster intervals remain negative, whereas the four SVM intervals include zero. This sensitivity check changes the resampling unit, not the primary test families or their multiplicity conclusions.

Fixed prompt-ensemble ablation

The released list contains 300 entries and 275 unique templates. All entries, including their released multiplicities, are retained. Clean accuracy is 92.90 ± 0.65%, and macro-F1 is 92.53 ± 0.59% across five folds. No prompts were selected using the test labels. The saved audio embeddings are bit-identical to the core run, and scores have been independently recomputed from the new unit text vectors. Thus the changed classifier output comes from the text references, not a new audio encoder or new training. The gain in clean accuracy does not remove the semantic-interference gap. Conditional capture rates must still be interpreted together with the unconditional capture fraction and total error count. The ensemble requires 15,000 text encodings to initialise its cache, compared with 50 for the single template. After averaging, both query paths score against 50 class vectors; one-off text preparation is excluded from query latency.

Table 9: Fixed prompt ensemble at +5 dB: five-fold macro-F1 (percent). The single-prompt reference is retained in Table 2.

Power	White	DEMAND	Same category	Different category
Own	75.72	84.32	55.50	53.69
Target	75.71	84.15	54.10	51.64

Table 10: CLAP error destinations at +5 dB under own-activity power. Each condition contains 2,000 trials. Captures are counts among all trials, not error-conditioned percentages.

	Single prompt		Fixed ensemble	
Condition	Errors	Captures	Errors	Captures
Same category	944	632	834	645
Different category	1015	612	876	645

The ensemble reduces other wrong-label predictions, while distractor-capture counts in these conditions are slightly higher. A larger fraction of the remaining errors therefore indicates a change in error composition as well as competition. Saved transition tables additionally distinguish new captures from predictions that already named the distractor in the gain-matched clean control. Correct-control-only rates use different subsets for different classifiers and are descriptive diagnostics rather than matched cross-model effects. An exploratory post-hoc sensitivity check gives each of the 275 unique templates equal weight. Across clean audio and all 24 interference cells, its largest absolute macro-F1 change is 0.47 percentage points. The declared 300-entry ensemble remains the reported intervention; this check is not used to select a better-performing prompt set. For high-overlap targets, the ensemble's own-protocol semantic-minus-matched-DEMAND accuracy differences are −31.56 points for same category and −34.13 points for different category. Their exploratory 95% target-source bootstrap intervals are $[-34.36, -28.84]$ and $[-36.77, -31.42]$. The target-activity protocol gives similar contrasts (−31.28 and −33.92 points). Thus improved clean readout does not eliminate the conditional semantic-interference gap; the gap against the ordinary-background control is larger than in the single-prompt reference.

Supervised augmentation transfer

All nine CNN variants (three regimes by three seeds) and all three SVM regimes are retained. The following tables include clean performance as well as transfer to each held-out interference condition. CNN entries summarise three separate five-fold means; their SD measures optimisation variation, not uncertainty from 15 independent test sets. SVM SD describes the five official folds. Paired accuracy effects are tested separately for each trained variant, with all 156 declared +5-dB intervention contrasts in one Holm family. Full paired intervals, discordance counts and gain-matched control scores accompany the exported data. The analysis was amended after initial WGN results to add direct semantic-versus- WGN contrasts answering the stated second question, before any semantic-trained result was available. The amendment changes the comparison family, not training settings or test mixtures; it is disclosed in the saved protocol.

Under own-activity power, WGN augmentation raises three-seed CNN white-noise macro-F1 from 7.48% to 43.95%, whereas same-category semantic performance changes only from 32.99% to 33.46%. Semantic training reaches 36.05% and 33.73% on same- and different-category mixtures, but its clean macro-F1 falls to 61.70%, compared with 65.78% for clean training and 64.65% for WGN training. Thus transfer is interference-specific, and the semantic intervention has a clean-performance cost. For CNN, 12 of 12 full-test-set semantic-condition comparisons of semantic versus WGN training have positive paired accuracy point estimates, but 0 of 12 pass the common 156-test Holm correction. For SVM, 4 of 4 full-test-set semantic-condition comparisons of semantic versus WGN training have positive paired accuracy point estimates, but 3 of 4 pass the common 156-test Holm correction. Here the 12 CNN comparisons are two semantic pairing types times two power protocols times three seeds; the four SVM comparisons omit the seed factor. White-noise

Table 11: Supervised intervention macro-F1 (percent): own-activity protocol at +5 dB. CNN: mean ± seed SD; SVM: mean ± fold SD.

Model	Training	Clean	White	DEMAND	Same	Different
SVM	Clean	47.96 ± 4.07	2.03 ± 0.86	16.13 ± 1.95	22.47 ± 2.68	21.34 ± 2.58
SVM	WGN	45.99 ± 4.06	41.14 ± 5.37	23.72 ± 5.08	22.60 ± 3.37	22.19 ± 2.58
SVM	Semantic	49.27 ± 2.39	1.91 ± 0.62	19.92 ± 2.57	26.83 ± 2.33	24.96 ± 0.91
CNN	Clean	65.78 ± 0.75	7.48 ± 0.42	30.92 ± 1.35	32.99 ± 0.34	31.41 ± 0.51
CNN	WGN	64.65 ± 0.61	43.95 ± 2.67	33.80 ± 2.19	33.46 ± 0.98	32.10 ± 1.05
CNN	Semantic	61.70 ± 2.40	9.91 ± 0.24	33.62 ± 1.55	36.05 ± 0.83	33.73 ± 0.86

Table 12: Supervised intervention macro-F1 (percent): target-activity protocol at +5 dB. CNN: mean ± seed SD; SVM: mean ± fold SD.

Model	Training	Clean	White	DEMAND	Same	Different
SVM	Clean	47.96 ± 4.07	2.03 ± 0.86	16.15 ± 2.29	21.14 ± 2.63	20.87 ± 2.37
SVM	WGN	45.99 ± 4.06	41.18 ± 5.43	23.76 ± 5.13	21.58 ± 3.21	20.65 ± 2.74
SVM	Semantic	49.27 ± 2.39	1.91 ± 0.62	19.61 ± 2.65	25.98 ± 2.05	24.13 ± 0.80
CNN	Clean	65.78 ± 0.75	7.48 ± 0.42	30.88 ± 1.43	31.99 ± 0.25	30.38 ± 0.49
CNN	WGN	64.65 ± 0.61	43.94 ± 2.69	33.83 ± 2.21	31.93 ± 0.87	30.86 ± 1.06
CNN	Semantic	61.70 ± 2.40	9.89 ± 0.21	33.68 ± 1.59	35.07 ± 0.92	32.53 ± 1.07

and DEMAND comparisons remain in the 156-test family but are not included in these semantic-condition counts. These tests concern per-variant paired accuracy, not significance of the three-seed mean macro-F1. The modest CNN advantage is therefore not established by the corrected tests.

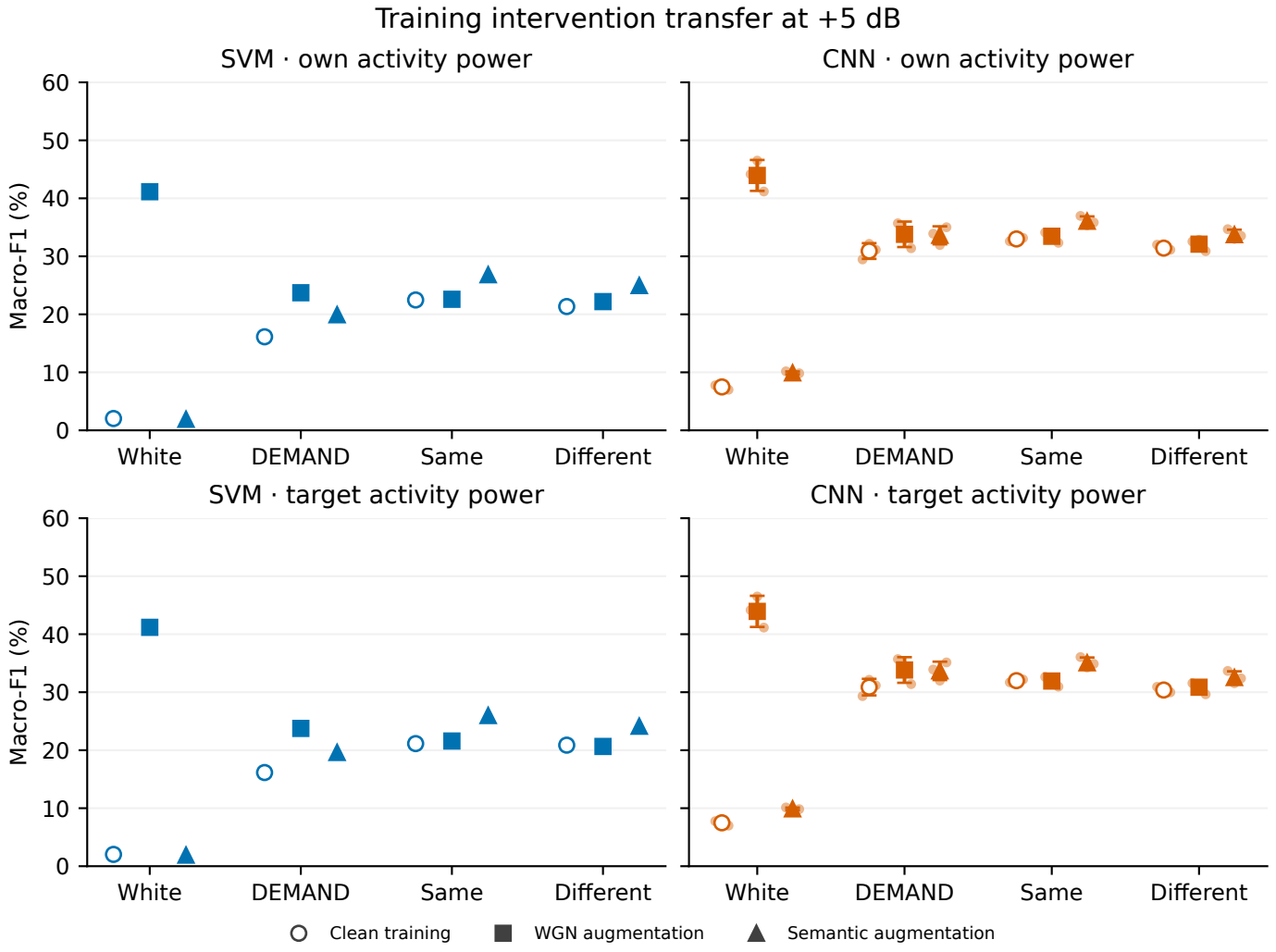


Figure 11: Augmentation transfer at +5 dB. Small CNN points retain each optimisation seed; large points and whiskers show the three-seed mean and sample SD. Each seed is a five-fold mean. SVM points are five-fold means; each cell retains 1,998–2,000 targets.

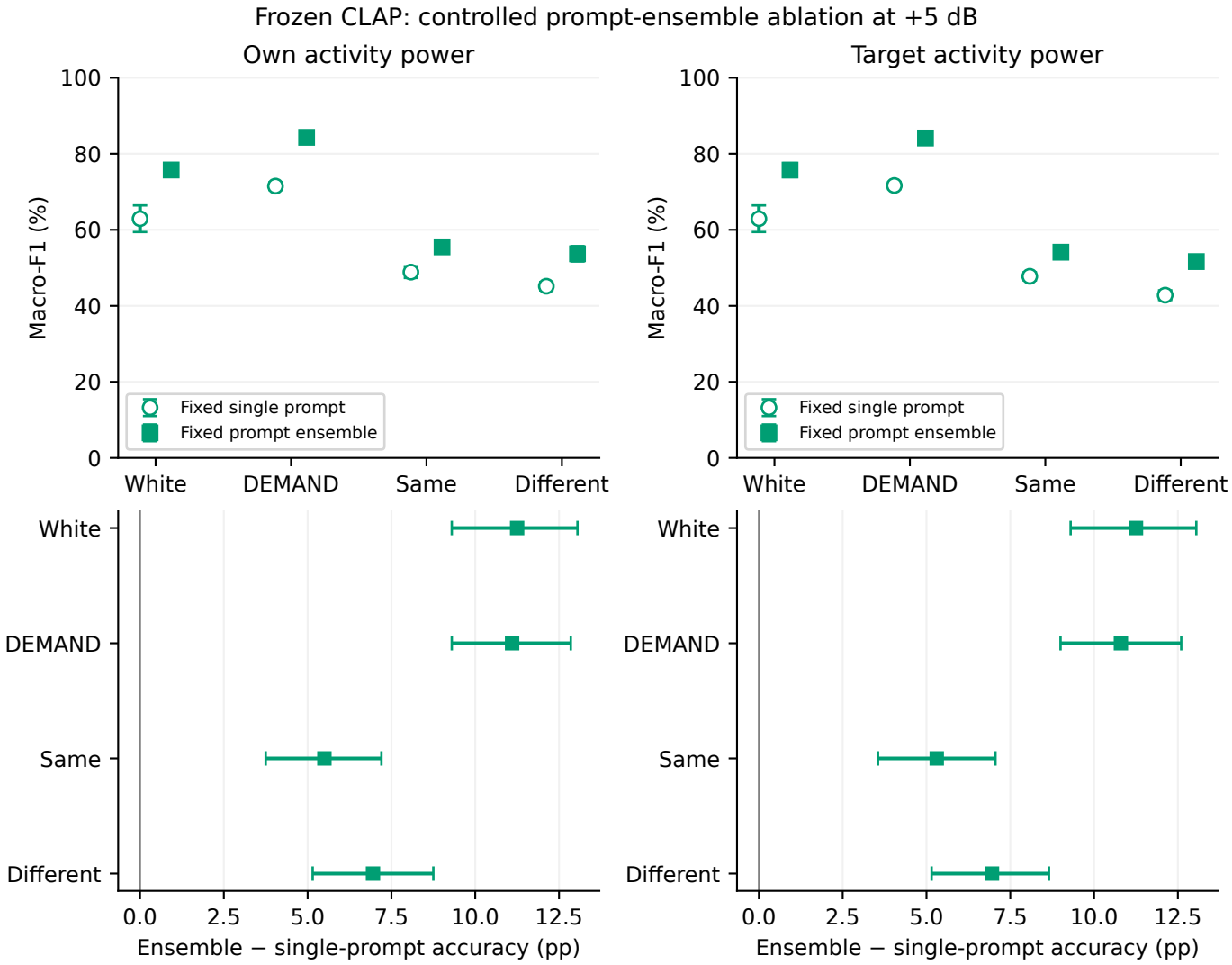


Figure 12: Frozen CLAP with identical audio vectors and two text readouts. Top: five-fold macro-F1 means with sample fold SD. Bottom: paired accuracy differences with pointwise 95% intervals from 2,000 within-fold target-bootstrap replicates, using 1,998–2,000 matched targets. These intervals are not multiplicity-adjusted; the associated exact tests belong to the 156-comparison Holm family.

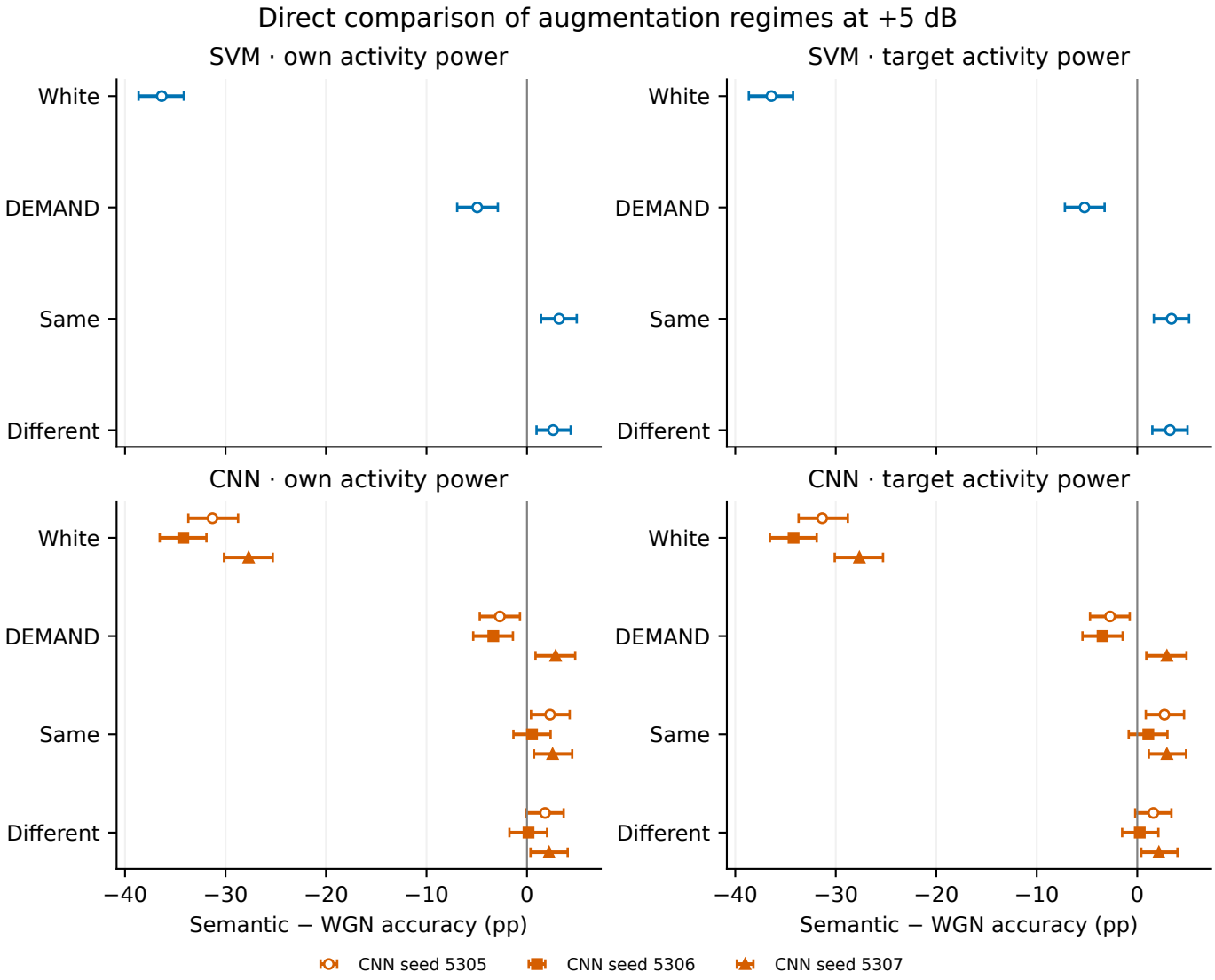


Figure 13: Direct semantic-augmentation minus WGN-augmentation accuracy effects at +5 dB. Each CNN comparison retains the same optimisation seed and 1,998–2,000 paired targets. Pointwise 95% intervals use 2,000 within-fold target-bootstrap replicates conditional on fitted models; exact tests join the common 156-comparison Holm family.

9 Completed scope and remaining limitations

The declared extension is complete: three clean CNN seeds, WGN and semantic training for SVM/CNN, fixed prompt ensembling, representation diagnostics and balanced deployment timing. The prompt experiment uses the released PAT template list, but does not implement PAT's full audio–text mutual alignment. Full PAT and Mix-CLAP are therefore not reported as reproduced methods. The two closing analyses—exact ranked-label readout and spectral–temporal overlap with DEMAND completion—are also complete. Experimental work is closed at this scope; no further models, datasets or training runs are proposed for this delivery.

Potential future studies, outside this project's completed scope, could reduce identified uncertainties. First, repeat held-out pairing with several fixed random seeds while keeping models frozen; report variability across distractor choices rather than assuming one pair per category type represents every competing scene. Second, construct and validate spectrally and temporally matched interferer controls with reduced recognisability. Recognisability must be checked rather than assumed from phase randomisation or another transformation; residual acoustic changes would still limit causal interpretation. Third, vary onset alignment for fixed source pairs and use a denser SIR grid on a declared subset, checking event truncation and achieved power after each shift. This would test overlap effects within pairs and support transition-threshold estimates that three SIR points cannot provide. The completed top-two diagnostic complements, but does not retrospectively replace, the designated-target metric. A calibrated detector or a dataset with exhaustive multi-event annotations would be a different evaluation task.

10 Conclusion

Interference power alone does not describe environmental-sound classifier failure. The SVM is vulnerable to white-noise-induced feature change, while CLAP shows substantial competing-event capture despite strong background-noise performance. Both power protocols and matched overlap strata are needed to interpret that pattern. The CNN contrast is directionally consistent across three seeds in the high-overlap population; the original seed's corrected test remains inconclusive. The smaller acoustic-closeness subsets do not reproduce a negative CNN contrast, whereas CLAP retains a substantial negative contrast. This population-dependent evidence rules out a simple explanation by nominal power alone, without isolating semantics causally.

Training interventions reveal a practical trade-off. WGN augmentation strongly protects the CNN against white noise but transfers little to semantic mixtures. Semantic augmentation gives larger semantic point-estimate gains while reducing clean CNN performance; its direct paired accuracy advantages over WGN training remain uncertain after the declared correction. Prompt ensembling improves CLAP's class readout without removing competing-event failures. Exact top-two readout recovers the target rank in some original captures, but many mixtures still exclude it from the returned pair. The project therefore connects controlled DSP, representation diagnostics and deployment cost with an explicit account of which robustness gains transfer and which do not.

Reproducibility and attribution

The repository contains custom DSP, supervised baselines, deterministic mixers, training/evaluation splits, score and embedding exports, analysis scripts and figure data. LAION-CLAP, its checkpoint and the PAT prompt list are external resources with explicit provenance. Model/data hashes and source snapshots identify the executed versions. The original proposal is retained, and its power definition change is documented above. AI-assisted implementation, checking and draft preparation were used; the report and code remain subject to author review and the course's disclosure requirements. All declared experimental runs are complete and independently checked. Final recording and course submission remain separate from these research outputs.

References

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[10] N. A. Schuster et al., "Noncollapsibility and its role in quantifying confounding bias in logistic regression," BMC Medical Research Methodology, 2021. doi:10.1186/s12874-021-01316-8.

A Supplementary figures

The following figures retain the original within-model scales and denominators. The repository also contains a complete corresponding figure set for the target power convention, each with numerical source tables and export audits.

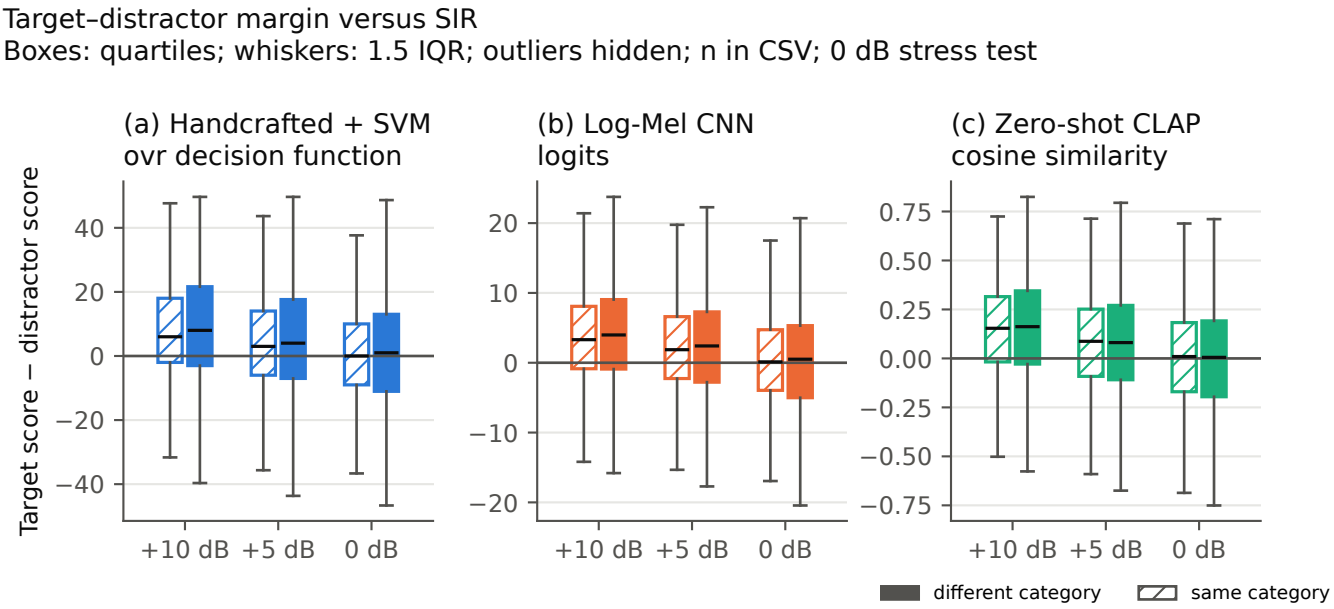


Figure 14: Target–distractor margin distributions. Scores have architecture-specific scales; comparisons are within each panel. A negative margin indicates a distractor score above the target score, not necessarily the overall predicted class.

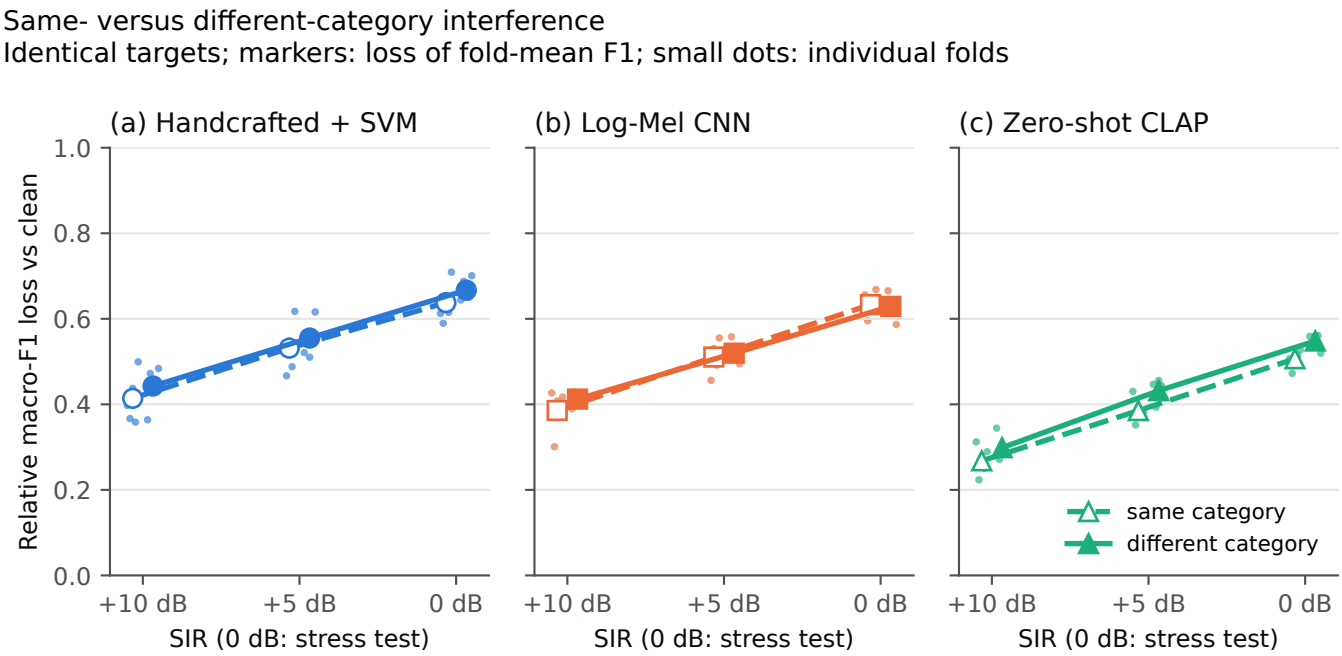


Figure 15: Same- versus different-category relative macro-F1 loss. Large markers are ratios of five-fold mean F1 values on common targets; small dots are the five individual fold ratios. A ratio of means is not generally the mean of ratios. Category relationships are coarse dataset groupings, not a validated perceptual-distance scale.

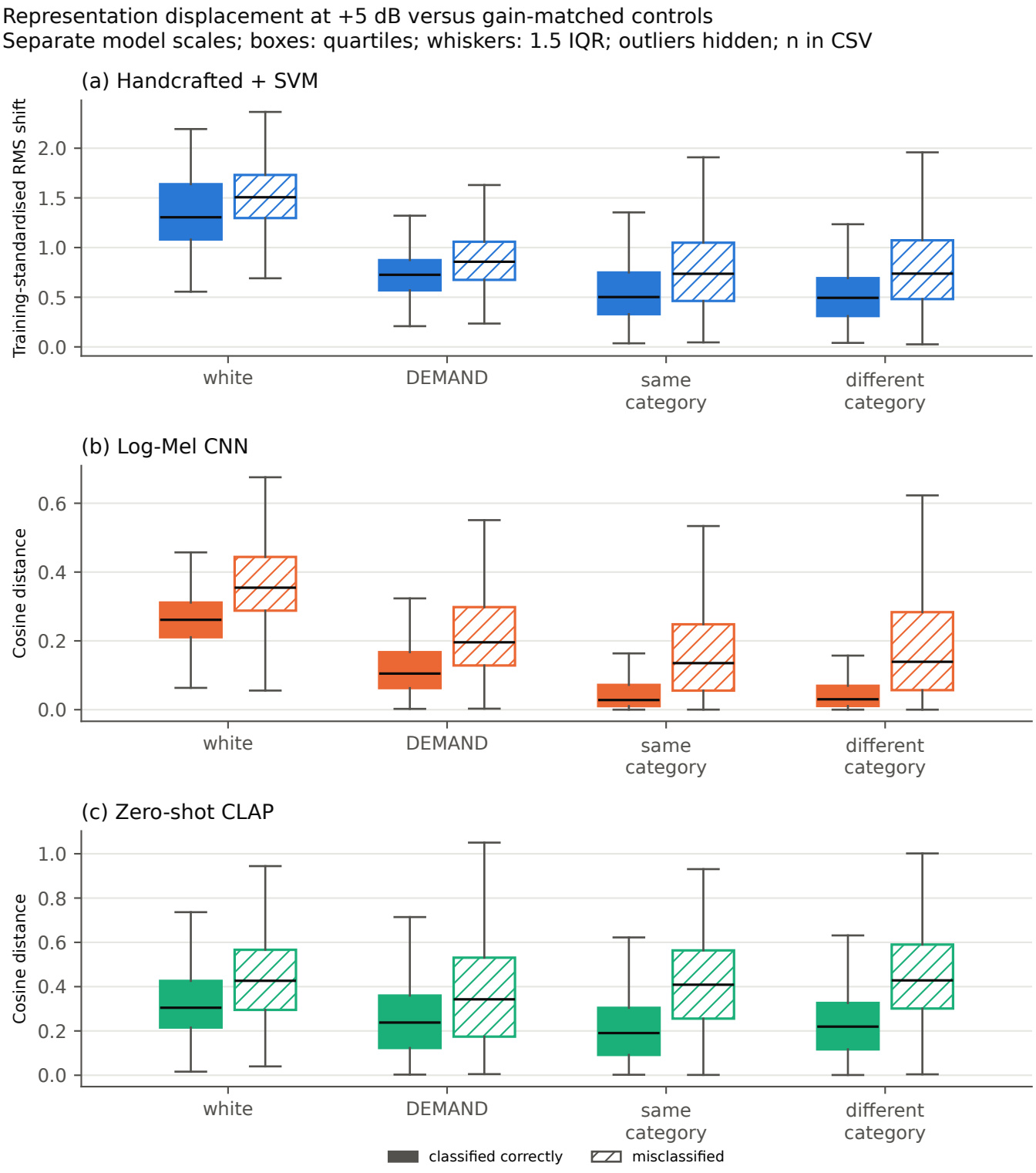


Figure 16: Representation displacement from gain-matched controls, separated by classification outcome. Handcrafted features use training-fold standardisation; CNN and CLAP use cosine distance. Absolute distances are not comparable between architectures.

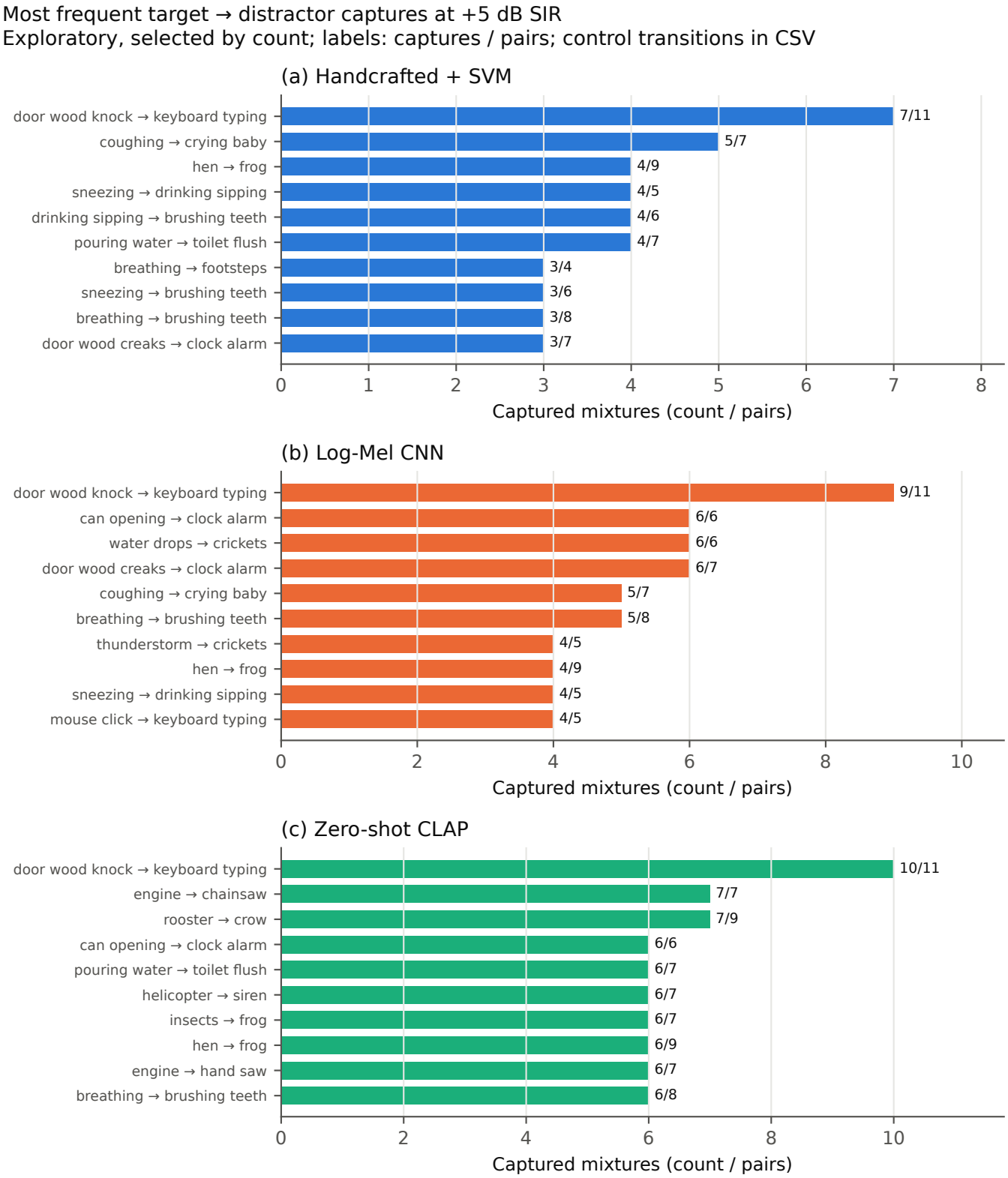


Figure 17: Observed target-to-distractor capture counts relative to available pairs. Counts include pre-existing clean-control confusions; they are not all newly induced errors. The source table separately counts captures from correct controls. Examples are selected by observed counts, and small denominators do not establish reliable class-pair risk.

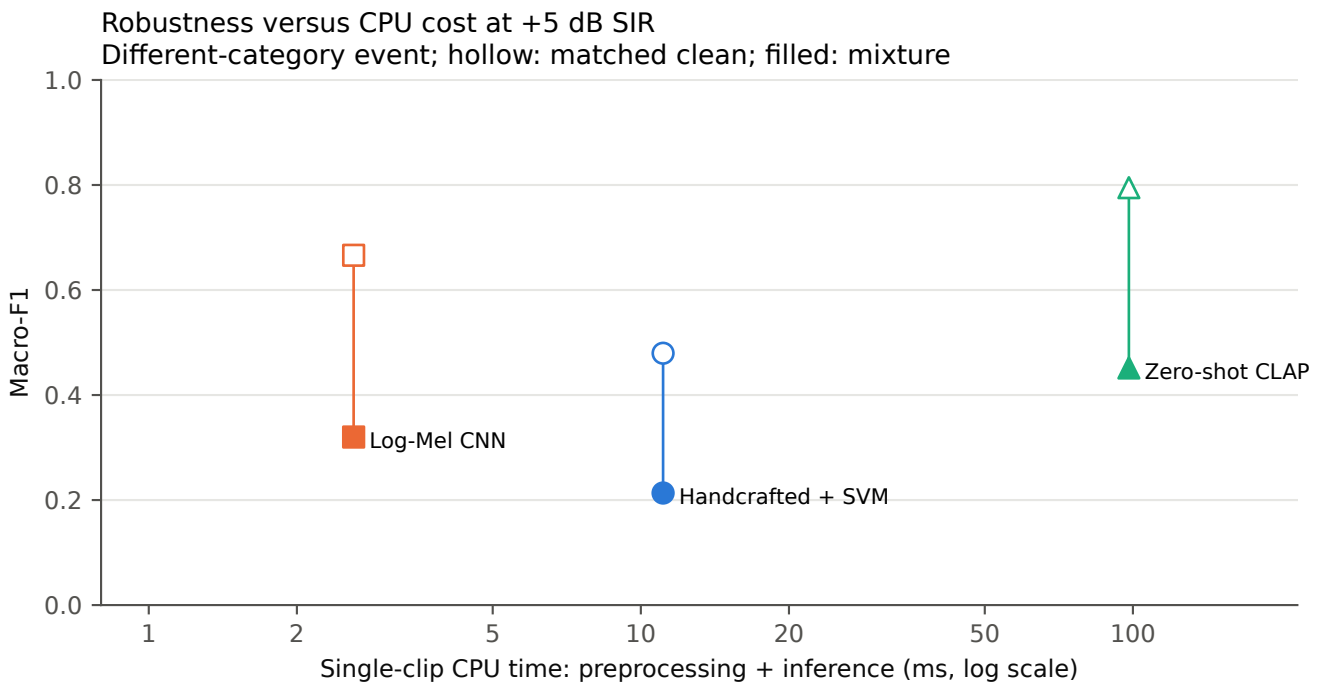


Figure 18: Original own-protocol robustness and CPU microbenchmark at +5 dB different-category interference. The horizontal coordinate sums the median preprocessing and median inference times for one clip; it is not the median total over the balanced panel. Cached CLAP text embeddings are used; open markers show clean macro-F1. External pretraining cost is excluded.

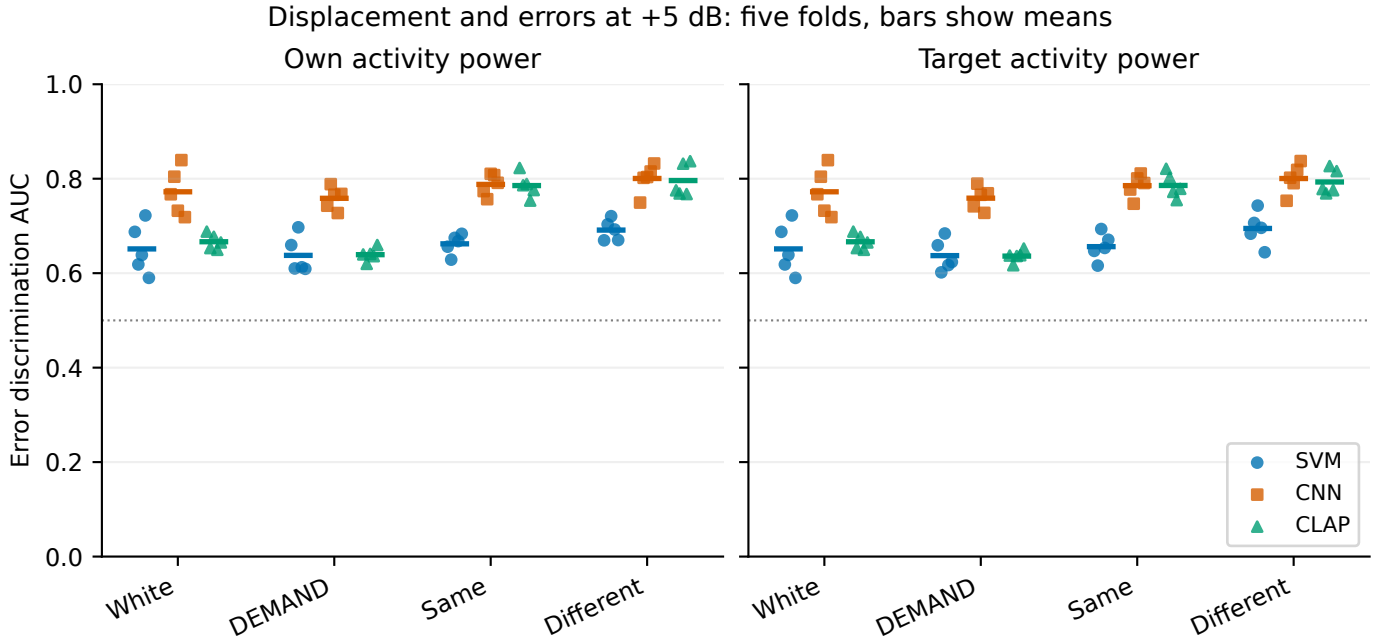


Figure 19: Association between representation displacement and classification error at +5 dB. Dots show the five official folds and short bars their means. AUC above 0.5 indicates larger movement tends to accompany errors; this is descriptive, not a causal semantic-effect estimate.

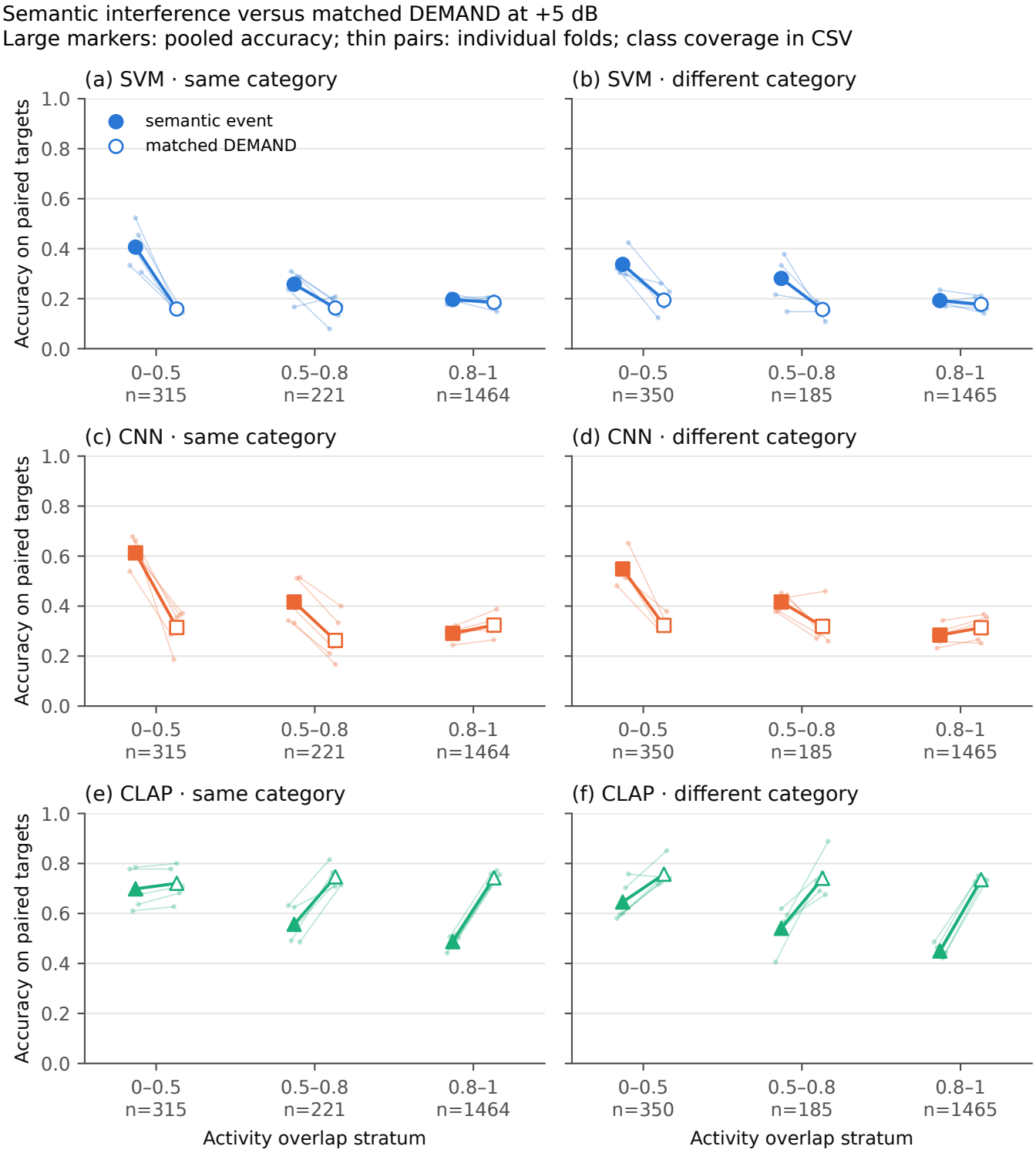


Figure 20: Overlap-stratified semantic versus DEMAND comparisons on matched target sets. Read accuracy together with class coverage; macro-F1 with all 50 labels is sensitive to absent classes in small strata.

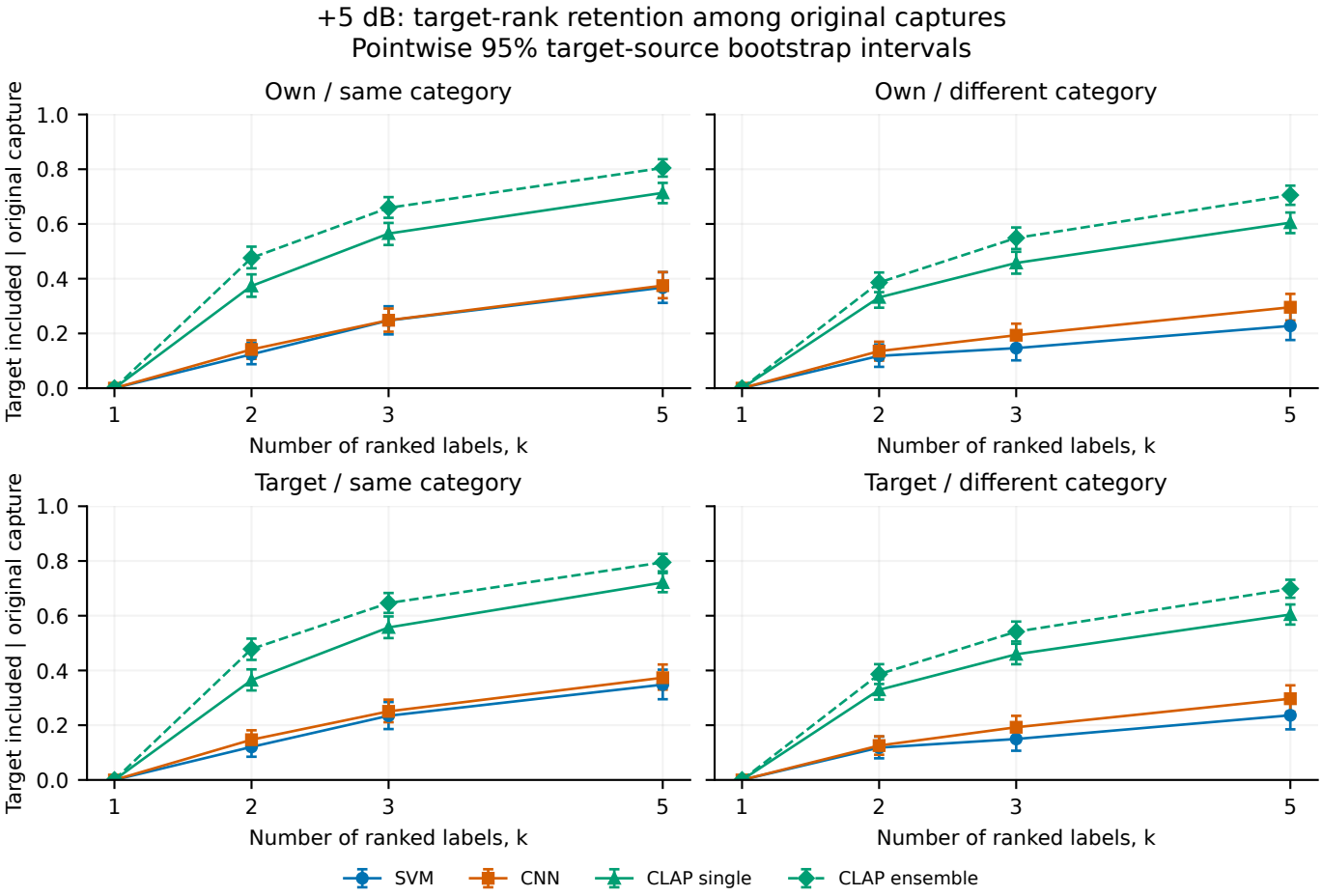


Figure 21: Target inclusion among original argmax captures at +5 dB as the number of returned labels increases. Pointwise 95% intervals resample original target-source clusters (2,000 replicates); repeated clips remain together. The target cannot be first in an argmax capture, so the $k = 1$ value is zero by definition. Larger k mechanically relaxes retrieval.

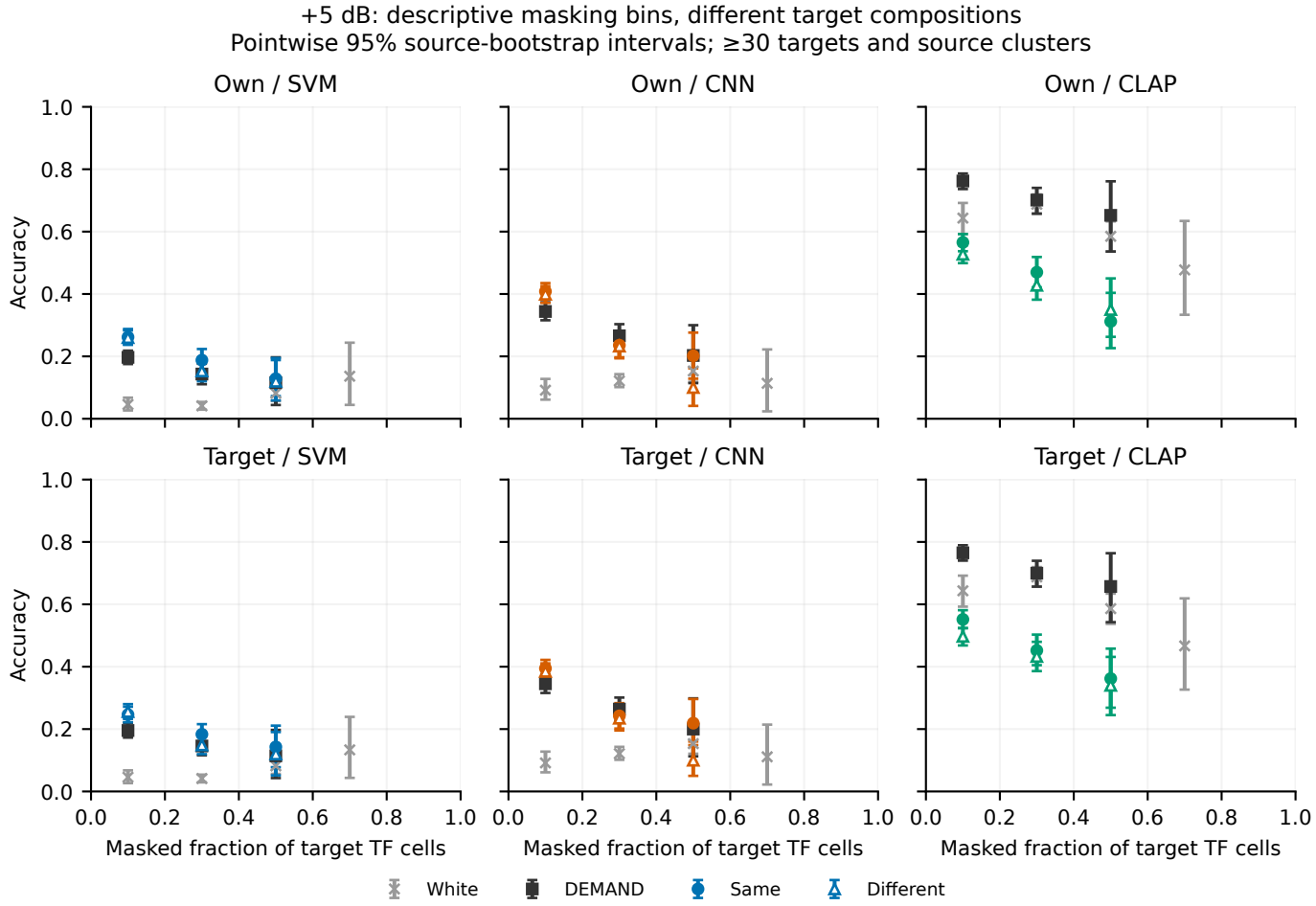


Figure 22: Accuracy inside fixed time–frequency masking bins at +5 dB. Points and source-bootstrap intervals require at least 30 targets and source clusters. Condition groups inside a bin can contain different targets and classes; this display is descriptive, unlike the same-target comparisons in Figure 8. Empty or small bins are not interpolated.